CHAPTER FOUR

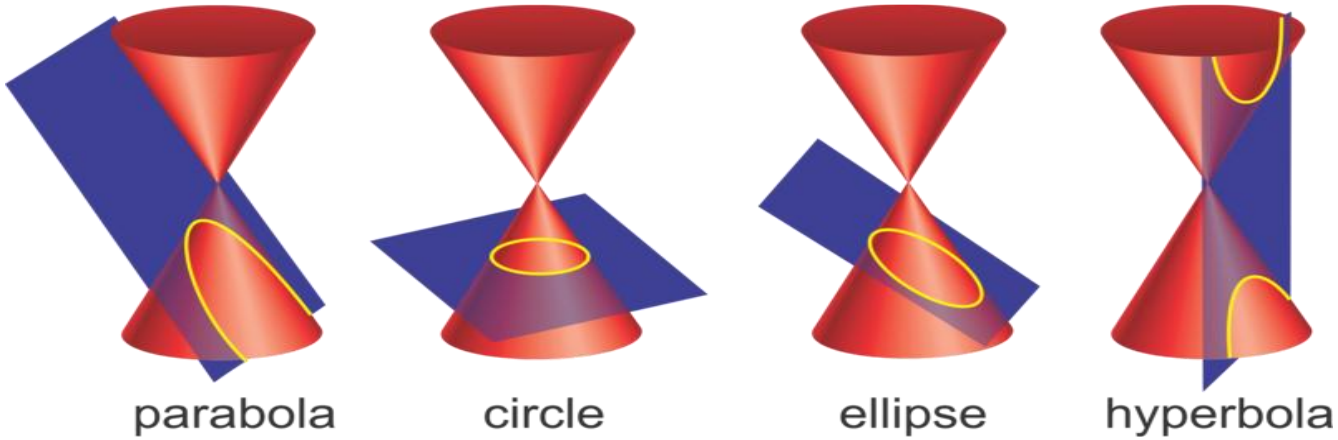
ANALYTICAL FUNCTIONS

The four shapes:

- parabolas,
- Circles,
- Ellipses, and
- Hyperbolas

are called **Conic sections.**

These shapes are created when a plane cuts a right circular double cone.



❖ Degenerate Conics

- When the plane does intersect the vertex of the cone, the resulting conic is called a degenerate conic.
- Degenerate conics include a point, a line, and two intersecting lines.
- The equation of every conic can be written in the following form:

$$\mathbf{Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0,}$$

where A, B, C, D, E and F are coefficients/constants.

4.1 Distance Formula and Equation of Lines

4.1.1 Distance between two points and division of segments

- If P and Q are two points on the coordinate plane, then PQ represents the line segment joining P and Q and $d(\mathrm{P},\mathrm{Q})$ or $|\mathrm{PQ}|$ represents the distance between P and Q.
- Recall that the distance between points a and b on a number line is $|a-b| = |b-a|$.
- Thus, the distance between two points $\mathrm{P}(x_1, y_1)$ and $\mathrm{R}(x_2, y_1)$ on a horizontal line must be $|x_2 - x_1|$ and the distance between $\mathrm{Q}(x_2, y_2)$ and $\mathrm{R}(x_2, y_1)$ on a vertical line must be $|y_2 - y_1|$.

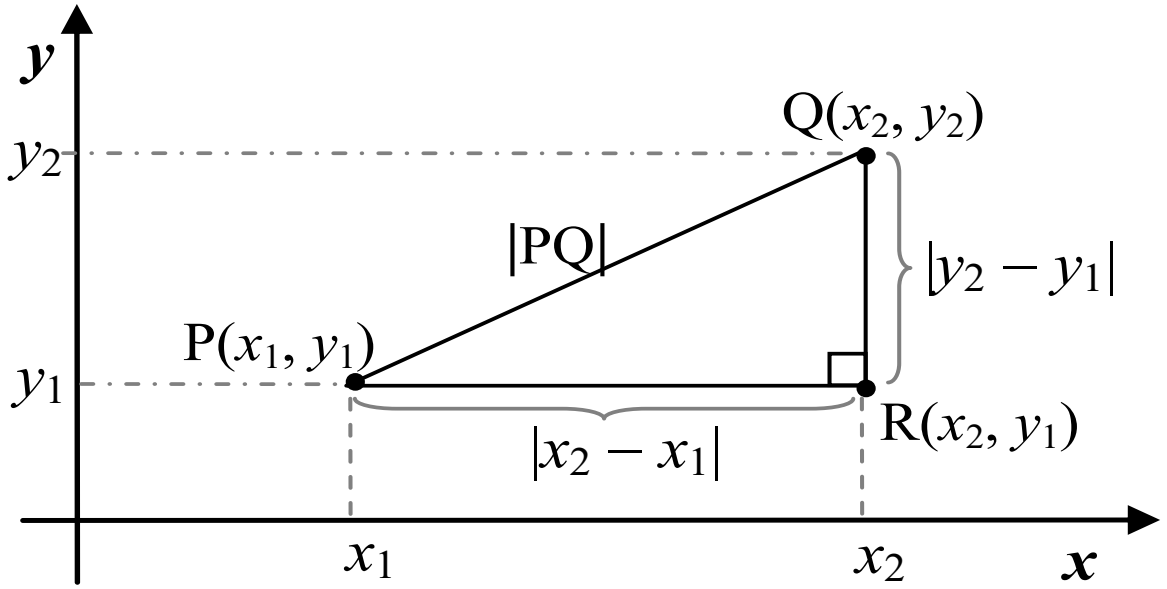


Figure 4.2

To find distance |PQ| between any two points $P(x_1, y_1)$ and $Q(x_2, y_2)$, we note that ΔPRQ in Figure 4.2 is a right triangle, and so by Pythagorean Theorem we get:

$$|PQ|^2 = |x_2 - x_1|^2 + |y_2 - y_1|^2 \Leftrightarrow |PQ| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Therefore, we have the following:

Distance Formula: The distance between the two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by:

$$|PQ| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Note that, from the distance formula, the distance between the origin $O(0,0)$ and a point $P(x, y)$ is

$$|OP| = \sqrt{x^2 + y^2}$$

Example 4.1: (i) The distance between $O(0,0)$ and $P(3,4)$ is

$$|OP| = \sqrt{3^2 + 4^2} = 5.$$

(ii) The distance between $P(1,2)$ and $Q(3,6)$ is

$$|PQ| = \sqrt{(3-1)^2 + (6-2)^2} = \sqrt{20}.$$

(iii) The distance between $P(-1,2)$ and $Q(5,-6)$ is

$$|PQ| = \sqrt{(5+1)^2 + (-6-2)^2} = 10.$$

Division point of a line segment: Given two distinct points P(x_1, y_1) and Q(x_2, y_2) in the coordinate plane, we want to find the coordinates (x_0, y_0) of the point R that lies on the segment PQ and divides the segment in the ratio r_1 to r_2; that is

$$\frac{|\mathrm{PR}|}{|\mathrm{RQ}|} = \frac{r_1}{r_2},$$

where r_1 and r_2 are given positive numbers. (See Figure 4.3).

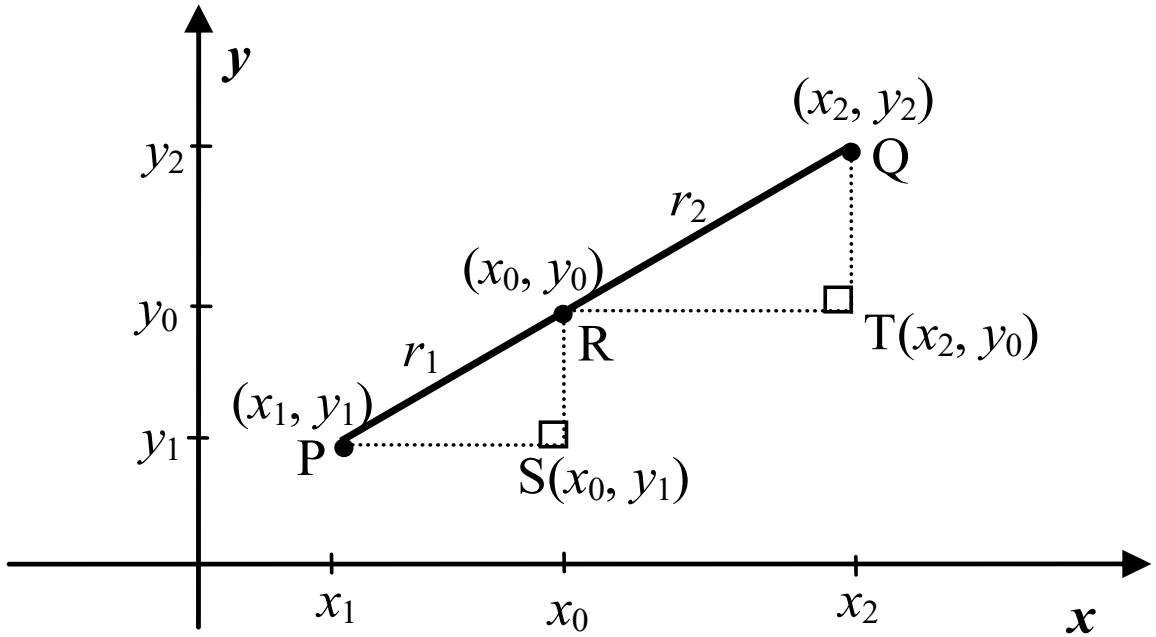


Figure 4.3

To determine (x_0, y_0), we construct two right triangles ΔPSR and ΔRTQ as in Figure 4.3. We then have $|PS| = x_0 - x_1$, $|SR| = y_0 - y_1$, $|RT| = x_2 - x_0$, and $|TQ| = y_2 - y_0$. Now since ΔPSR is similar to ΔRTQ, we have that

$$\frac{x_0 - x_1}{x_2 - x_0} = \frac{r_1}{r_2} \quad \text{and} \quad \frac{y_0 - y_1}{y_2 - y_0} = \frac{r_1}{r_2}$$

or $r_2(x_0 - x_1) = r_1(x_2 - x_0)$ and $r_2(y_0 - y_1) = r_1(y_2 - y_0)$.

Solving for x_0 and y_0, we obtain $x_0 = \dfrac{x_1 r_2 + x_2 r_1}{r_1 + r_2}$ and $y_0 = \dfrac{y_1 r_2 + y_2 r_1}{r_1 + r_2}$

Theorem 4.1: Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be distinct points in the coordinate plane. If $R(x_0, y_0)$ is a point on the line segment PQ that divides the segment in the ratio $|PR| : |RQ| = r_1 : r_2$, then the coordinates of R is given by

$$(x_0, y_0) = \left(\frac{x_1 r_2 + x_2 r_1}{r_1 + r_2}, \frac{y_1 r_2 + y_2 r_1}{r_1 + r_2} \right)$$

In particular, the midpoint of PQ is given by $\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2} \right)$

Example 4.2: Given P(−3, 3) and Q(7,8),

(i) find the coordinates of the point R on the line segment PQ such that $|PR| : |RQ| = 2 : 3$.

(ii) find the coordinates of the midpoint of PQ.

Solution: (i) Obviously $R(x_0, y_0)$ is given by

$$(x_0, y_0) = \left(\frac{-3\times 3 + 7\times 2}{2+3}, \frac{3\times 3 + 8\times 2}{2+3} \right) = (1, 5)$$

(ii) The coordinates of the midpoint is $\left(\frac{-3+7}{2}, \frac{3+8}{2} \right) = (2,\ 11/2)$.

4.1.2 Equations of lines

An equation of a line l must be satisfied by the coordinates (x, y) of every point on the line.

A line can be vertical, horizontal or oblique. The equation of a vertical line that intersects the x-axis at $(a, 0)$ is $x=a$ because the x-coordinate of every point on the line is a.

Similarly, the equation of a horizontal line that intersects the y-axis at $(0, b)$ is $y=b$ because the y-coordinate of every point on the line is b.

An oblique line is a straight line which is neither vertical nor horizontal.

To find equation of an oblique line we use its slope which is the measure of the steepness of the line. In particular, the slope of a line is defined as follows.

Definition 4.1. The **slope** of a non-vertical line that passes through the points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ is

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

The slope of a vertical line is not defined. Note that the slope of horizontal line is 0.

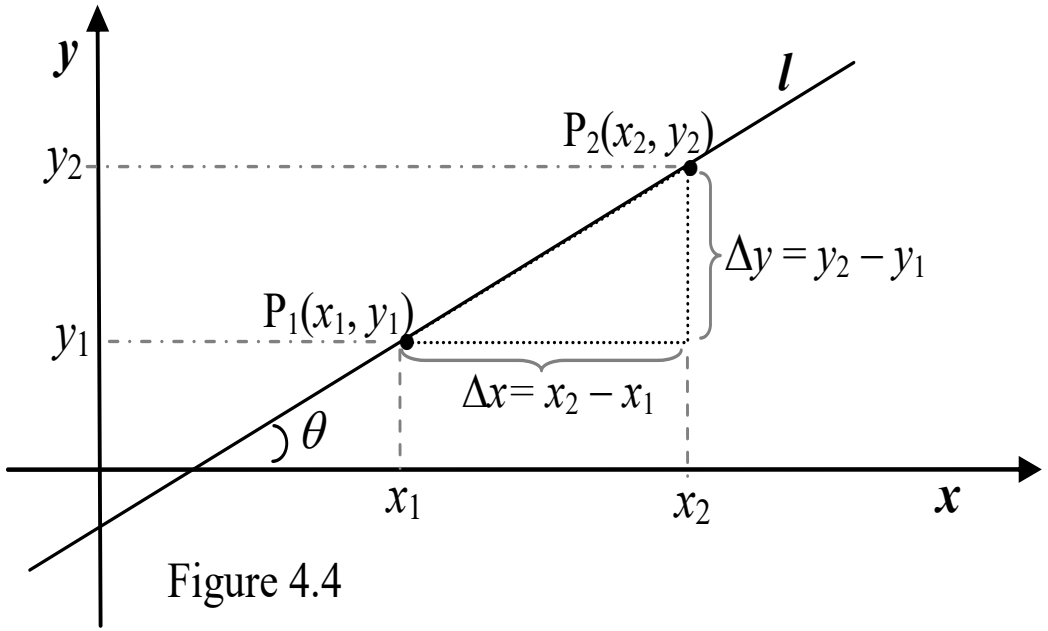


Figure 4.4

This leads to the following equation of the line:

$$y - y_1 = m(x - x_1)$$

(called point-slope form of equation of a line).

In general, depending on the given information, you can show that the equations of oblique lines can be obtained using the following formulas.

Given Information	Formula for Equation of the Line
Slope m and its y-intercept $(0,b)$	Slope-Intercept-Form: $y = mx + b$
Slope m and a point (x_1, y_1) on $\boldsymbol{l}$	Point-Slope-Form: $y - y_1 = m(x - x_1)$ Or $y = m(x - x_1) + y_1$
Two points (x_1, y_1) and (x_2, y_2) on $\boldsymbol{l}$	Two-Point Form: $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$
x-intercept $(a,0)$ and y-intercept $(0,b)$	Intercept Form: $\frac{x}{a} + \frac{y}{b} = 1$

Example 4.3: Find an equation of the line l if

(i) the line passes through $(3, -2)$ and its angle of inclination is $135°$.

(ii) the line passes through the points $(1, 2)$ and $(4, -2)$

Solution: (i) The slope of l is $m = \tan(135°) = -1$; and it passes through point $(3, -2)$. Th

using the point-slope form with $x_1 = 3$ and $y_1 = -2$, we obtain the equation of the line as

$y - (-2) = -1(x - 3)$ which simplifies to $y = -x + 1$.

(ii) Given the line passes through $(1, 2)$ and $(4, -2)$, the slope of the line is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-2-2}{4-1} = \frac{-4}{3}.$$

So, using the point-slope form with $x_1 = 1$ and $y_1 = 2$, we obtain the equation of the line as

$y - 2 = \frac{-4}{3}(x - 1)$ which simplifies to $4x + 3y = 10$.

General Form: In general, the equation of a straight line can be written as $ax + by + c = 0$, for constants a, b, c with a and b not both zero. Indeed, if $a = 0$ the line is a horizontal line given by $y = -\frac{c}{b}$, if $b = 0$ the line is a vertical line given by $x = -\frac{c}{a}$, and if both $a, b \neq 0$ it is the oblique line given by $y = -\left(\frac{a}{b}\right)x - \frac{c}{b}$ with slope $m = -\frac{a}{b}$ and y-intercept $-\frac{c}{b}$.

Parallel & Perpendicular lines: Slopes can be used to check whether lines are parallel, perpendicular or not. In particular, let $\boldsymbol{l}_1$ and $\boldsymbol{l}_2$ be non-vertical lines with slope m_1 and m_2, respectively. Then,

1. $\boldsymbol{l}_1$ and $\boldsymbol{l}_2$ are parallel, denoted by $\boldsymbol{l}_1 \,||\, \boldsymbol{l}_2$, iff $m_1 = m_2$.
2. $\boldsymbol{l}_1$ and $\boldsymbol{l}_2$ are perpendicular, denoted $\boldsymbol{l}_1 \perp \boldsymbol{l}_2$, iff $m_1 m_2 = -1$ (or $m_1 = -\frac{1}{m_2}$).

   Moreover, if $\boldsymbol{l_1}$ and $\boldsymbol{l_2}$ are both vertical lines then they are parallel. However, if one of them is horizontal and the other is vertical, then they are **perpendicular**.

Example 4.4: Find an equation of the line through the point $(3, 2)$ that is parallel to the line $2x + 3y + 5 = 0$.

Example 4.5: Show that the lines $2x + 3y + 5 = 0$ and $3x - 2y - 4 = 0$ are perpendicular.

4.1.3 Distance between a point and a line

Suppose a line l and a point P(x,y) not on the line are given. The distance from P to l, $d(\mathrm{P}, l)$, is defined as the perpendicular distance between P and l. That is,

$d(\mathrm{P}, l) = |\mathrm{PQ}|$, where Q is the point on l such that $\mathrm{PQ} \perp l$. (See Figure 4.5)

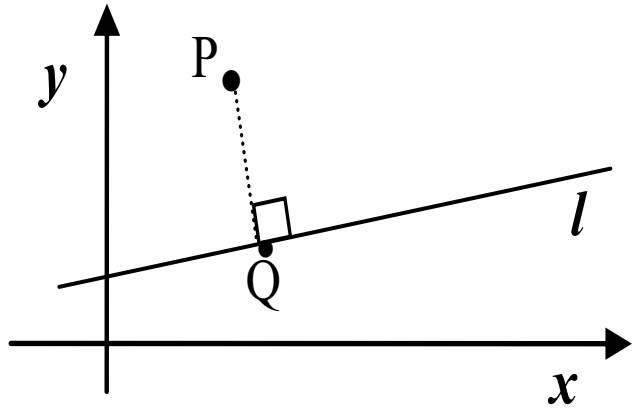


Figure 4.5: $|\mathrm{PQ}| = d(\mathrm{P}, l)$

If P is on l, then $d(\mathrm{P}, l) = 0$. Moreover, given a point P(h,k) observe that

(i) if the line l is a horizontal line $y=b$, then $d(\mathrm{P}, l) = |k - b|$.

(ii) if the line l is a vertical line $x=a$, then $d(\mathrm{P}, l) = |h - a|$

In general, however, to find the distance between a point $P(x_0, y_0)$ and an arbitrary line $\boldsymbol{l}$ given by $ax + by + c = 0$, we have to first get a point Q on $\boldsymbol{l}$ such that $PQ \perp \boldsymbol{l}$ and then compute $|PQ|$. This yields the formula given in the following Theorem.

Theorem 4.2: The distance between a point $P(x_0, y_0)$ and a line L: $ax + by + c = 0$ is given by

$$d(P, L) = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$$

In particular, if we take $(x_0, y_0)=(0,0)$ in this formula, we obtain the distance between the origin $O(0,0)$ and a line L : $ax + by + c = 0$ which is given by

$$d(O, L) = \frac{|c|}{\sqrt{a^2 + b^2}}$$

Example 4.6: Show that the origin and $P(6,4)$ are equidistant from the line $L: y = -(3/2)x + 13/2$.

Solution: By equidistant we mean equal distance.

So, we need to show $d(O,L) = d(P,L)$.

To use the above formula, we first write the equation of the line L in the general form which is $3x + 2y - 13 = 0$. Thus, $a = 3$, $b = 2$ and $c = -13$.

$$d(O,L) = \frac{|c|}{\sqrt{a^2+b^2}} = \frac{|-13|}{\sqrt{9+4}} = \frac{13}{\sqrt{13}} \quad \text{and}$$

$$d(P,L) = \frac{|a(6)+b(4)-13|}{\sqrt{a^2+b^2}} = \frac{|3\times6+2\times4-13|}{\sqrt{9+4}} = \frac{13}{\sqrt{13}}.$$

Therefore, $d(O,L) = d(P,L) = \frac{13}{\sqrt{13}}$.

Thus, O(0,0) and P(6,4) are equidistant from the given line L.

4.2 Circles

4.2.1 Definition of a Circle

Definition 4.2. A **circle** is the set of points that are equidistant from a fixed point called **center of a circle** and such an equidistant is called its **radius**.

A line segment that passes through the center and has endpoints on the circle is a **diameter**. Diagrammatically we have:

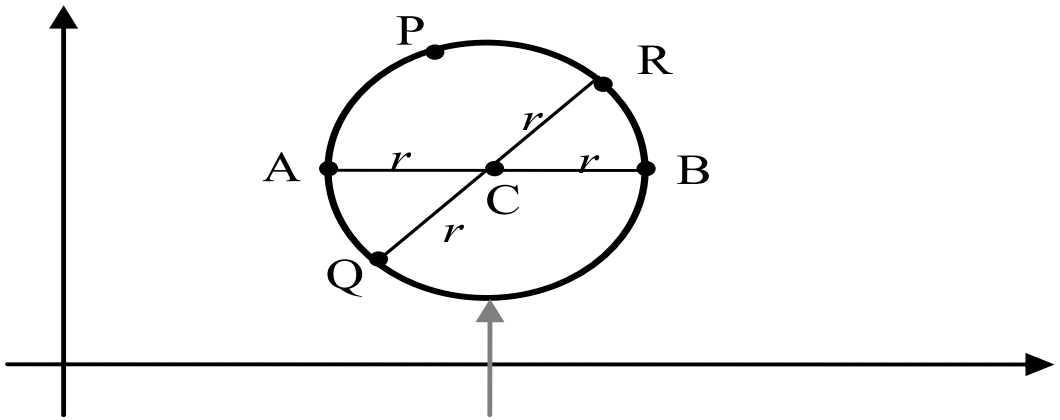


Figure 4.6. Circle with center C, radius r

Based on the definition:

- a point P is on the circle if and only if its distance to C is r, that is $|CP| = r$.
- point in the plane is said to be inside the circle if its distance to the center C is less than r.
- a point in the plane is said to be outside the circle if its distance to C is greater than r.

A chord of the circle is a line segment whose endpoints are on the circle.

A diameter is a chord of the circle through the center C.

Consequently, C is the midpoint of a diameter and the length of a diameter is $2r$.

Example 4.7: Consider a circle of radius 5 whose center is at $C(2,1)$. Determine whether each of the following points is on the circle, inside the circle or outside the circle: $P_1(5,5)$, $P_2(4,5)$, $P_3(-2,5)$, $P_4(-1,-2)$, $P_5(2,-4)$, $P_6(7,0)$.

4.2.2 Equation of a Circle

We now construct an equation that the coordinates (x,y) of the points on the circle should satisfy. So, let P(x,y) be any point on a circle of radius r and center C(h,k) (see, Figure 4.7). Then, the definition of a circle requires that

$$|\text{CP}| = r$$

$$\Rightarrow \quad \sqrt{(x-h)^2+(y-k)^2} = r$$

or

$$\boxed{(x-h)^2+(y-k)^2 = r^2}$$

(Standard equation of a circle with center (h,k) and radius r.

In particular, if the center is at origin, i.e., $(h,k) = (0,0)$, the equation is

$$\boxed{x^2+y^2 = r^2}$$

(Standard Equation of a circle of radius r centered at origin)

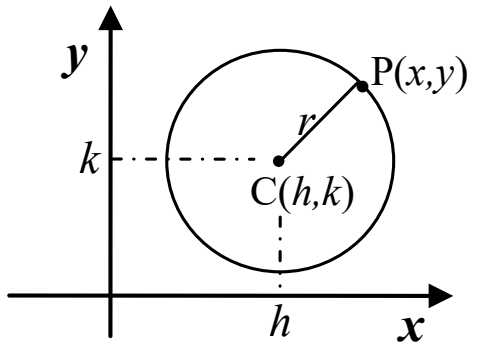
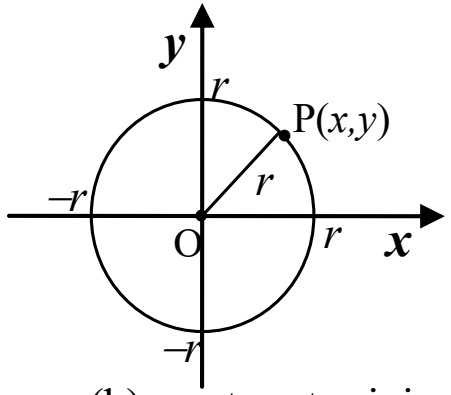


Figure 4.7 circles (a) center at C(h,k)



(b) center at origin

Example 4.8: Find an equation of the circle with radius 4 and center $(-2, 1)$.

Solution: Using the standard equation of a circle in which the center $(h, k) = (-2, 1)$ and radius $r = 4$ we obtain the equation

$$(x+2)^2 + (y-1)^2 = 16$$

Example 4.9: Find the equation of a circle with endpoints of a diameter at P(–2, 0) and Q(4, 2).

Solution: The center of the circle $C(h,k)$ is the mid-point of the diameter. Hence,

$(h,k)=\left(\frac{-2+4}{2},\frac{0+2}{2}\right)=(1,1)$. Also, for its radius r, $r^2=|CP|^2=(1+2)^2+(1-0)^2=10$.

Thus, the equation of the circle is $(x-h)^2+(y-k)^2=r^2$. That is,

$$(x-1)^2+(y-1)^2=10.$$

Example 4.10: Suppose P(−2,4) and Q(5,3) are points on a circle whose center is on x-axis. Find the equation of the circle.

Solution: We need to obtain the center C and radius r of the circle to construct its equation. As the center is on x-axis, its second coordinate is 0. Therefore, let $C(h,0)$ be the center of the circle. Note that $|PC|^2 = |QC|^2 = r^2$ as both P and Q are on the circle. So, from the first equality we get $(-2-h)^2 + 4^2 = (5-h)^2 + 3^2$. Solving this for h we get $h=1$. Hence, the center is at $C(1, 0)$ and $r^2 = |QC|^2 = (5-1)^2 + 3^2 = 25$. Therefore, the equation of the circle is

$(x-1)^2 + y^2 = 25$.

Example 4.11: Determine whether the given equation represents a circle. If it does, identify its center and radius and sketch its graph.

(a) $x^2 + y^2 + 2x - 6y + 7 = 0$

(b) $x^2 + y^2 + 2x - 6y + 10 = 0$

(c) $x^2 + y^2 + 2x - 6y + 11 = 0$

Solution: We need to rewrite each equation in standard form to identify its center and radius. We do this by completing the square on the x-terms and y-terms of the equation as follows:

(a) $(x^2 + 2x) + (y^2 - 6y) = -7.$ (Grouping x-terms and y-terms)

$\Leftrightarrow (x^2 + 2x + 1^2) + (y^2 - 6y + 3^2) = -7 + 1 + 9.$ (Adding 1^2 and 3^2 to both sides)

$\Leftrightarrow \quad (x+1)^2 + (y-3)^2 = 3.$

Comparing this with the standard equation of circle this is equation of a circle with center $(h, k) = (-1, 3)$ and radius $r = \sqrt{3}$. The graph of the circle is sketched in Figure 4.8

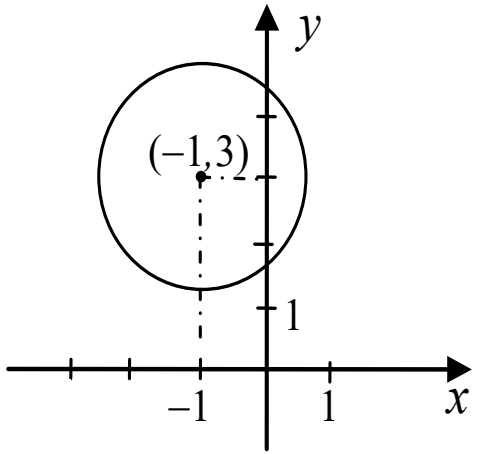


Figure 4. 8

(b) Following the same steps as in (a), you can see that $x^2 + y^2 + 2x - 6y + 10 = 0$ is equivalent to $(x+1)^2 + (y-3)^2 = 0$.

This is satisfied by the point $(-1, 3)$ only. The locus of this equation is considered as a point-circle, circle of zero radius (sometimes called degenerated circle).

(c) Again following the same steps as in (a), you can see that $x^2 + y^2 + 2x - 6y + 11 = 0$ is equivalent to $(x+1)^2 + (y-3)^2 = -1$.

Note that this does not represent a circle; in fact it has no locus at all (Why?).

Remark: Consider an equation of the form

$$x^2 + y^2 + Dx + Ey + F = 0.$$

By completing the square you can show the following:

- If $D^2 + E^2 - 4F > 0$, then the equation represents a circle with center $\left(-\frac{D}{2}, -\frac{E}{2}\right)$ and radius $r = \frac{1}{2}\sqrt{D^2+E^2-4F}$.
- If $D^2 + E^2 - 4F = 0$, then the equation is satisfied by the point $\left(-\frac{D}{2}, -\frac{E}{2}\right)$ only. In this case the locus of the equation is called **point-circle** (circle of zero radius).
- If $D^2 + E^2 - 4F < 0$, then the equation has no locus.

4.2.3 Intersection of a circle with a line and tangent line to a circle

The number of intersection points of a given line and a circle is at most **two**; that is, either no intersection point, or only one intersection point, or two intersection points.

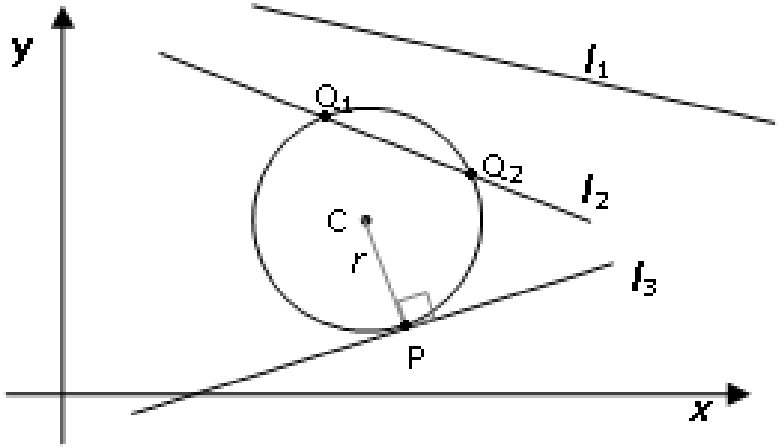


Figure 4.9: Intersection of a line and circle

For instance, in Figure 4.9, the line l_1 has no intersection with the circle, l_2 has two intersection points with the circle, namely, Q_1 and Q_2, and l_3 has only one intersection point with the circle, namely, P.

A line which intersects a circle at one and only one point is called a **tangent line** to the circle. In this case, the intersection point is called the **point of tangency**. Thus, l_3 a tangent line to the circle in Figure 4.9 and P is the point of tangency.

In general, given a circle of radius *r* with center $C(h,k)$ and a line $\boldsymbol{l}$, by computing the distance $d(C,\boldsymbol{l})$ between C and $\boldsymbol{l}$ we can conclude the following.

1. If $d(C,\boldsymbol{l}) > r$, then the line does not intersect with the circle.
2. If $d(C,\boldsymbol{l}) < r$, then the line is a secant of the circle; that is, they have two intersection points.
3. If $d(C,\boldsymbol{l}) = r$, then $\boldsymbol{l}$ is a tangent line to the circle. The point of tangency is the point P on the line (and on the circle) such that $CP \perp \boldsymbol{l}$. This means the product of the slopes of $\boldsymbol{l}$ and CP must be -1.

Example 4.12 Write the equation of the circle tangent to the ***x***-axis at (6,0) whose center is on the line $x - 2y = 0$.

Solution: The circle in the question is as in Figure 4.10.

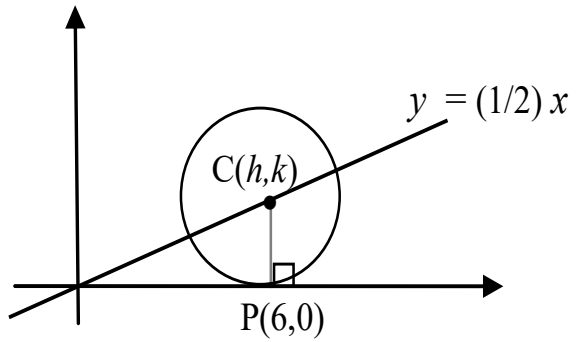


Figure 4.10

Let C(h, k) be the center of the circle. (h, k) is on the line $y = (1/2)x \Rightarrow k = (1/2)h$; and the circle is tangent to x-axis at P(6,0) $\Rightarrow$ CP should be perpendicular to the x-axis.

$\Rightarrow h = 6 \Rightarrow k = 3$ and the radius is $r = |CP| = k - 0 = 3$.

Hence, the circle is centered at (6, 3) with radius $r = 3$. Therefore, the equation of the circle is $(x-6)^2 + (y-3)^2 = 9$.

Example 4.13 Suppose the line $y=x$ is tangent to a circle at point P(2,2). If the center of the circle is on the x-axis, then what is the equation of the circle?

Solution: The circle in the question is as in Figure 4.11.

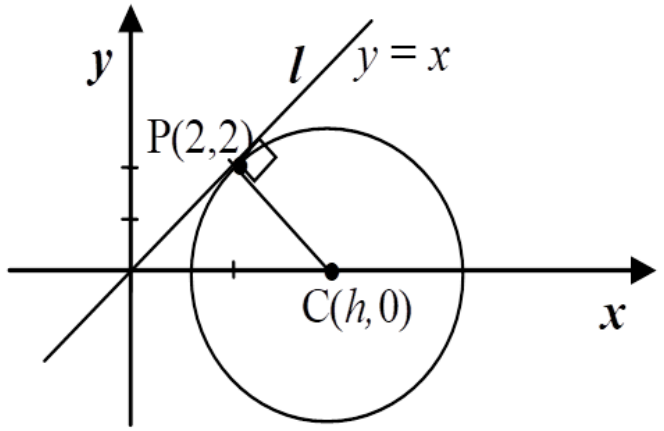


Figure 4.11

Let the center of the circle be C(h,0). We need to find h. The slope of the line $l: y = x$ is 1 and l is perpendicular to CP. Hence the slope of CP is –1.

So, the slope of $CP = \frac{2-0}{2-h} = -1 \Longrightarrow h = 4.$

The center of the circle is $C(4,0)$; and

$$r^2 = |\text{CP}|^2 = (2-0)^2 + (2-4)^2 = 4 + 4 = 8.$$

Therefore, the equation of the circle is $(x-4)^2 + y^2 = 8.$

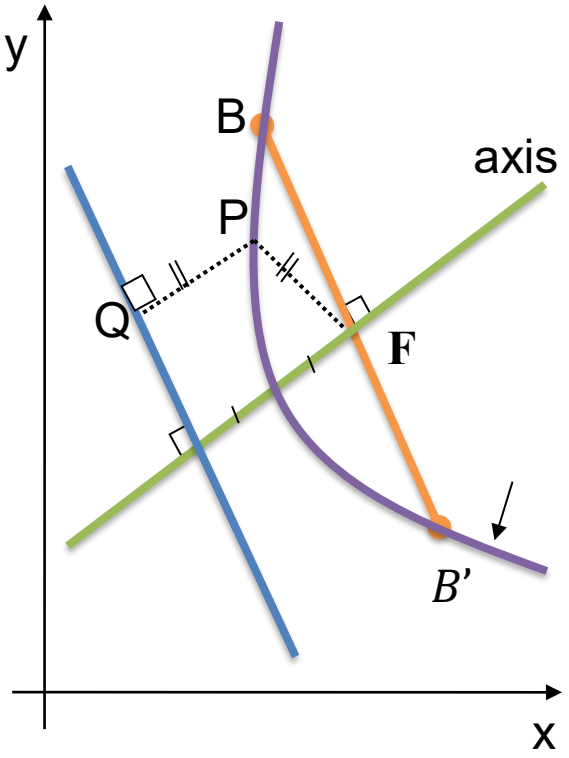
4.3 Parabolas

4.3.1 Definition of a Parabola

Definition 4.3: Let **L** be a fixed line and **F** be a fixed point not on the line, both lying on a plane. A parabola is a set of points equidistant from **L** and **F**.

In this case,

- the line **L** is called the **directrix** and
- the fixed point **F** is called the **focus** of the parabola.



- Note that the point halfway between the focus F & directrix L is on the parabola; it is called the **vertex,** denoted by V.
- |VF| is called the **focal length**.
- The line through F perpendicular to the directrix is called the **axis** of the parabola and it is the line of symmetry for the parabola.
- The chord BB′ through F perpendicular to the axis is called **latus rectum**.
- The length of the latus rectum, i.e, |BB′|, is called **focal width**.
- Let $|VF| = p$, then we have $|BB'| = 4p$

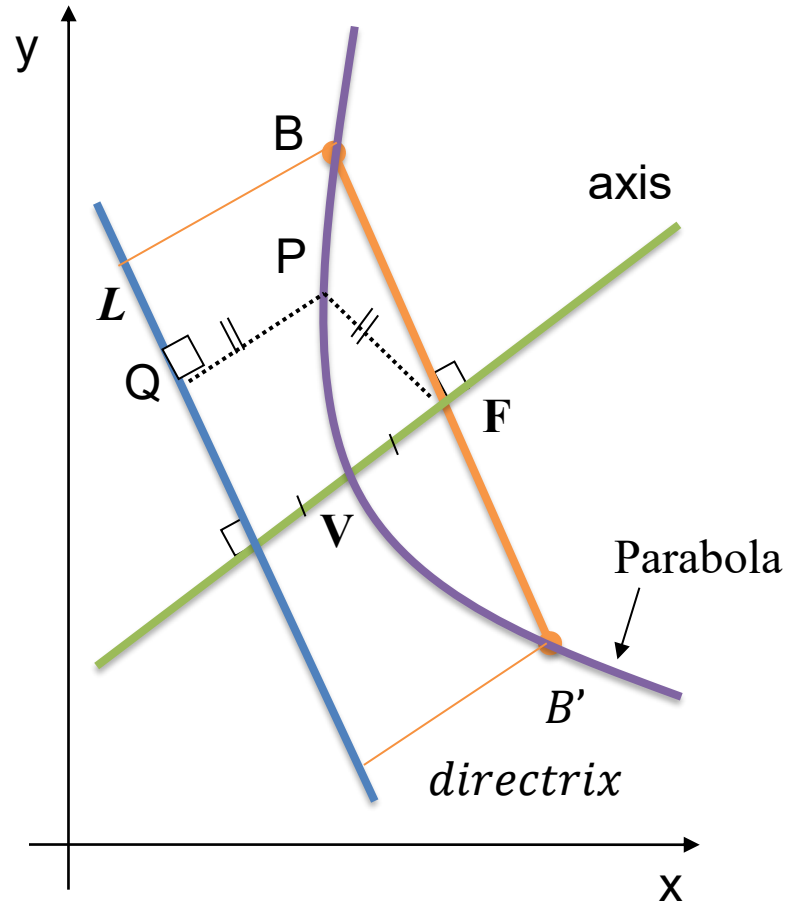


Figure 4.12: Parabola, $d(\mathrm{P},L) = |\mathrm{PF}|$

4.3.2 Equation of Parabolas

I: Equation of a parabola whose axis is parallel to the *y*-axis:

A parabola whose axis is parallel to ***y***-axis is called **vertical parabola**.

A vertical parabola is either open upward (as in Figure 4.13 (a)) or open downward (as in Figure 4.13 (b)).

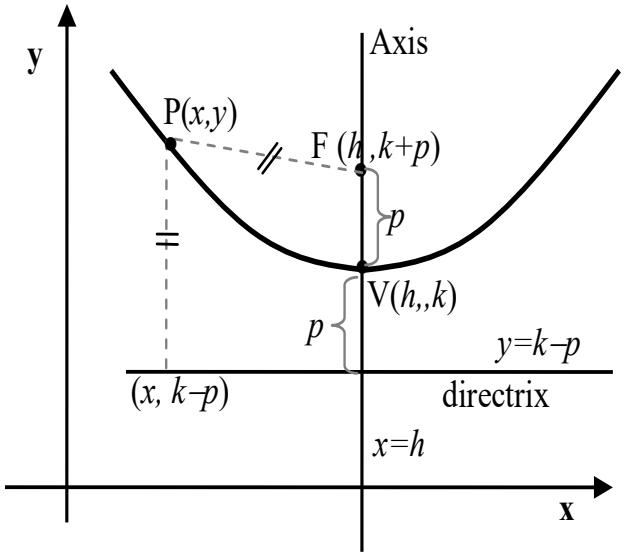
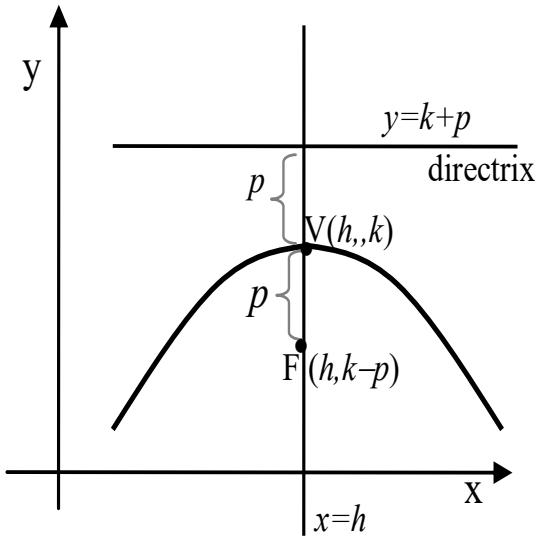


Figure 4.13: (a) parabola open upward



(b) parabola open downward

Considering a vertical parabola with vertex V(h,k) that opens upward (Figure 4.13a), its focus is at $F(h, k+p)$.

$\Rightarrow$ The equation of its directrix is $y = k-p$.

Then, for any point P(x,y) on the parabola, |PF| is equal to the distance between P and the directrix if and only if

$$\sqrt{(x-h)^2+(y-k-p)^2} = y-k+p$$

Upon simplification, this becomes

$$(x-h)^2 = 4p(y-k)$$

called standard equation of a vertical parabola,

vertex (h, k), focal length p, open upward.

In particular, if the vertex of a vertical parabola is at origin, i.e., $(h, k)=(0,0)$ and opens upward, then its equation is

$$x^2 = 4py$$

(In this case, its focus is at F(0, p), and its directrix is $y=-p$)

If a vertical parabola with vertex V(h, k) opens downward, then its directrix is above the parabola and its focus lies below the vertex (see Figure 4.13(b).

In this case the focus is at F(h, $k-p$), and its directrix is given by $y=k+p$. Moreover, following the same steps as above , the equation of this parabola becomes

$$(x-h)^2 = -4p(y-k)$$

(Standard equation of a vertical parabola, open downward, vertex (h, k) , and focal length p .)

In particular, if the vertex of a vertical parabola is at origin, i.e., (h, k) =(0,0) and opens downward, then its equation is

$$x^2 = -4py$$

(In this case, its focus is at F(0,$-p$), and its directrix is $y = p$)

Example 4.14: Find the vertex, focal length, focus and directrix of the parabola $y = x^2$.

Solution: The given equation, $x^2 = y$, is the standard equation of the parabola with vertex at origin (0,0) and $4p = 1 \Rightarrow$ its focal length is $p = 1/4$. Since the parabola opens upward, its focus is p units above its vertex $\Rightarrow$ its focus is at F(0,1/4); and its directrix is horizontal line p units below its vertex $\Rightarrow$ its directrix is $y = -1/4$. You may sketch this parabola.

Example 4.15: If a parabola opens upward and the endpoints of its latus rectum are at $A(-4, 1)$ and $B(2, 1)$, then find the equation of the parabola, its directrix and sketch it.

Solution: Since the focus F of the parabola is at the midpoint of its latus rectum AB, we have

$F=\left(\frac{-4+2}{2},\frac{1+1}{2}\right)=(-1,1)$, and focal width $4p=|AB|=2-(-4)=6 \Rightarrow$ focal length $p = 3/2$.

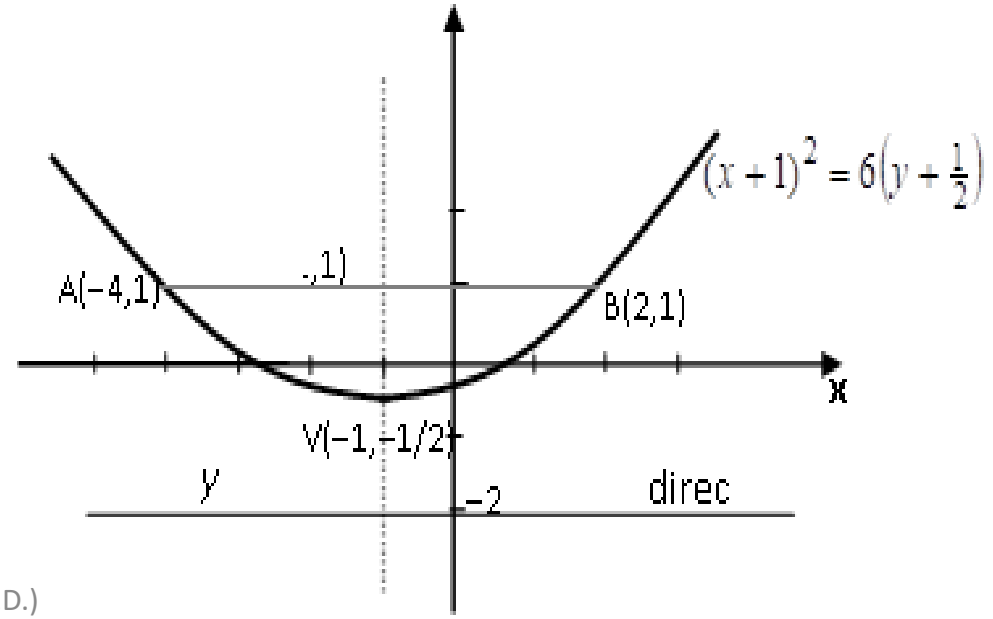
Moreover, as the parabola opens upward its vertex is p units below its focus. That is,

V (h, k) = (−1, 1−3/2) = (−1, −1/2). Therefore, the equation of the parabola is

$$(x+1)^2=6\left(y+\frac{1}{2}\right).$$

And its directrix is horizontal line p units below its vertex, which is $y = -1/2 -3/2 = -2$.

The parabola is sketch in the Figure 4.14 .



II: Equation of a parabola whose axis is parallel to the *x*-axis.

A parabola whose axis is parallel to ***x***-axis is called **horizontal parabola**. Such parabola opens either to the right or to the left as shown in Figure 4.15 (a) and (b), respectively.

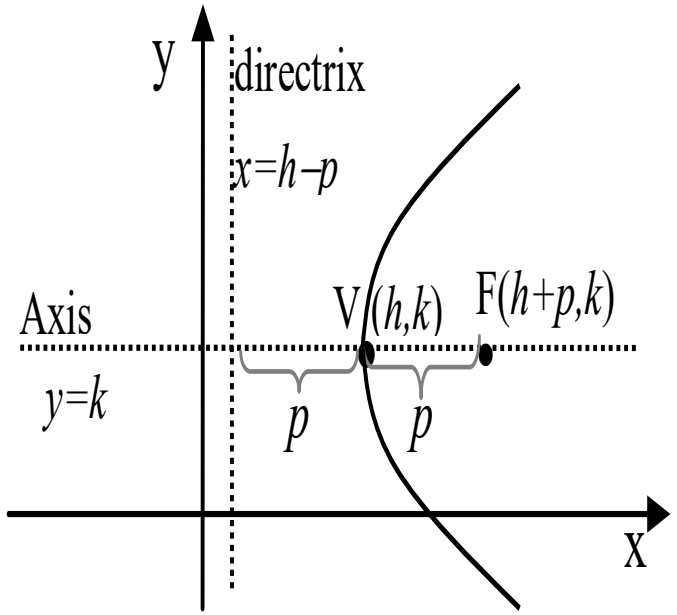
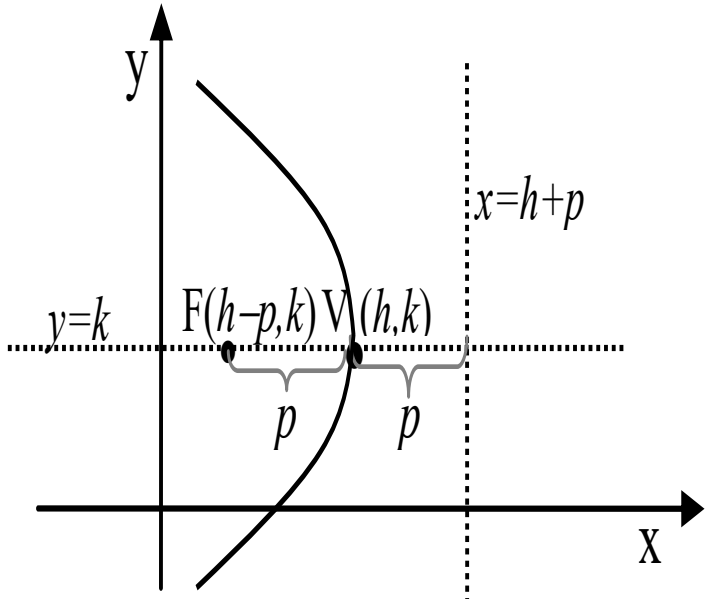


Figure 4.15: (a) Parabola open to the right



(b) Parabola open to the left

The equations of such parabolas can be obtained by interchanging the role of x and y in the equations of the parabolas discussed previously. These equations are stated below. In both cases, let the vertex of the parabola be at $V(h,k)$.

- If a horizontal parabola **opens to the right (**as in Fig.4.15(a)), then its focus is to the right of V at $F(h+p, k)$, its directrix is $x = h-p$, and its equation is

$$(y-k)^2 = 4p(x-h)$$

- If a parabola **opens to the left (**as in Figure 4.15 (b)), then its focus is to the left of V at $F(h-p,k)$, its directrix is $x=h+p$, and its equation is:

$$(y-k)^2 = -4p(x-h)$$

If the vertices of these parabolas are at the origin (0,0), then you can obtain their corresponding equations by setting $h=0$ and $k=0$ in the above equations.

Example 4.16: Find the focus and directrix of the parabola $y^2 + 10x = 0$ and sketch its graph.

Solution: The equation is $y^2 = -10x$; and comparing this with the above equation, it is an equation of a parabola whose vertex is at (0,0), axis of symmetry is the x-axis, open to the left and $4p=10$, i.e., $p=5/2$. Thus, the focus is F=(–5/2,0) and its directrix is $x = 5/2$. Its graph is sketched in Figure 4.16.

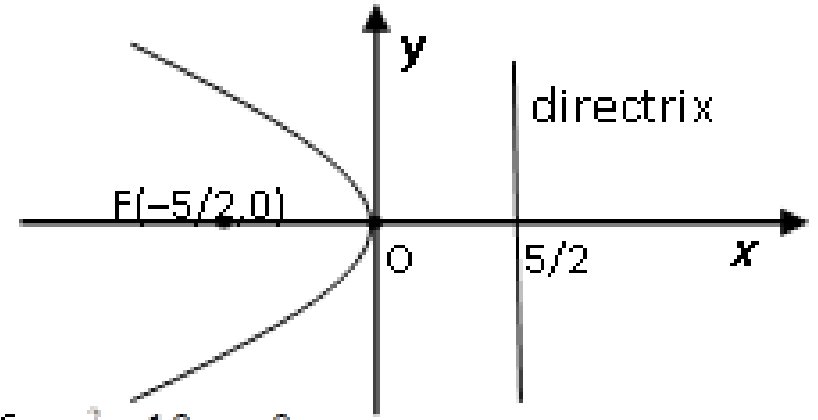


Figure 4.16: $y^2 + 10x = 0$

Example 4.17: Find the focus and directrix of the parabola $y^2 + 4y + 8x - 4 = 0$ and sketch it.

Solution: The eaquation is $y^2+4y=-8x+4$. (Now complete the square of y-terms)

$$\Rightarrow y^2+4y+\mathbf{2^2}=-8x+4+\mathbf{4}$$

$$\Rightarrow (y+2)^2=-8x+8$$

$$\Rightarrow (y+2)^2=-8(x-1)$$

This is equation of a parabola with vertex at $(h, k)=(1,-2)$, open to the left and focal length p, where $4p=8 \Rightarrow p=2$. Therefore, its focus is $F=(h-p, k)=(-1,-2)$, and directrix $x=h+p=3$.

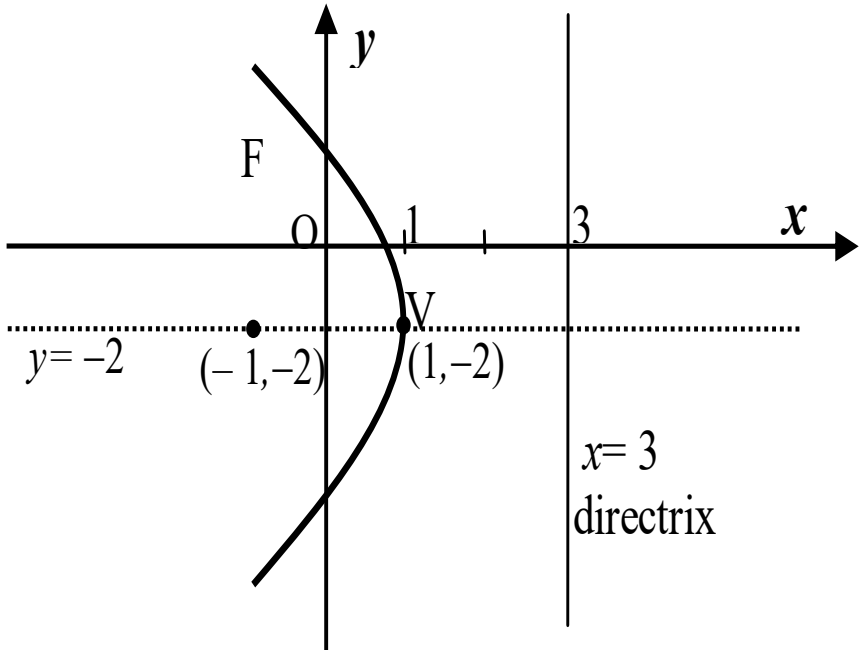


Figure 4.17: $y^2+4y+8x-4=0$

Remark:- An equation given as: $Ax^2 + Dx + Ey + F = 0$

or $Cy^2 + Dx + Ey + F = 0$

may represent a parabola whose axis is parallel to the ***y***-axis or parallel to the ***x***-axis, respectively. The vertex, focal length and focus for such parabolas can be identified after converting the equations into one of the standard forms by completing the square.

$$-y^2 + 6y - 8x + 31 = 0$$

$$-x^2 + 8x - 12y - 4 = 0$$

4.4 Ellipse

4.4.1 Definition of an Ellipse

Definition 4.4: Let F and F′ be two fixed points in the plane. An **ellipse** is the locus or set of all points in the plane such that the sum of the distances from each point to F and F′ is constant. That is, a point P is on the ellipse if and only if $|PF| + |PF'| = \text{constant}$. (See Figure 4.18).

The two fixed points, F and F′, are called **foci** (singular- **focus**) of the ellipse.

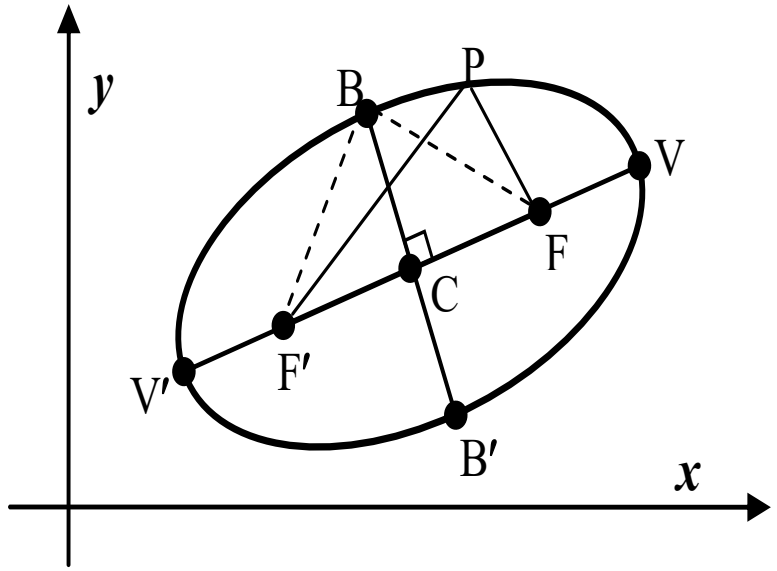


Figure 4.18: Ellipse: $|PF| + |PF'| = \text{constant}$

Note also the following terminologies and relationships about ellipse.

- The midpoint C between the foci F′ and F is called the **center** of the ellipse.
- The longest diameter (longest chord) V′V through F′ and F is called the **major axis** of the ellipse; and the chord BB′ through C which is perpendicular to V′V is called **minor axis**.
- The endpoints of the major axis, V′ and V, are called the **vertices** of the ellipse.

▪ From the definition, $|V'F'| + |V'F| = |VF'| + |VF| \Rightarrow |V'F'| = |VF| \Rightarrow |CV'| = |CV|$. Hence, C is the midpoint of V′V. We denote the length of the major axis by $2a$. That is, $|CV| = a$.

$\Rightarrow |VF'| + |VF| = |V'V| = 2a$.

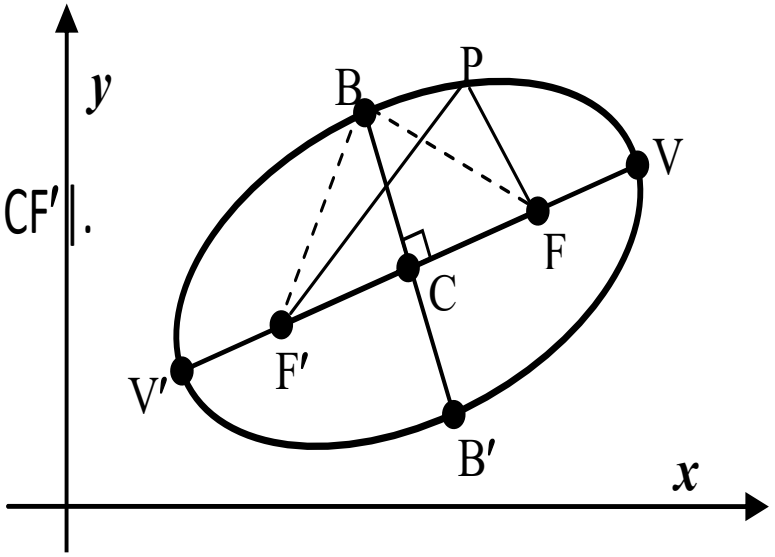
$\Rightarrow |PF'| + |PF| = 2a$, for any point P on the ellipse.

▪ We let $|BC| = b$. (You can show that C is the midpoint of BB′. So, $|B'C| = b$.)

▪ The distance from the center C to a focus F (or F′) is denoted by c, i.e., $|CF| = c = |CF'|$.

▪ Now, since $|BF'| + |BF| = 2a$ and BC is a perpendicular bisector of F′F, we obtain that $|BF'| = |BF| = a$. Hence, using the Pythagoras Theorem on ΔBCF, we obtain

$$b^2 + c^2 = a^2 \quad \text{or} \quad b^2 = a^2 - c^2.$$



- The ratio of the distance between the two foci to the length of the major axis is called the **eccentricity** of the ellipse, and denoted by e. That is,

$$e = \frac{|F'F|}{|V'V|} = \frac{c}{a}. \qquad \text{(Note that } 0 < e < 1 \text{ because } 0 < c < a \text{)}$$

4.4.2 Equation of an Ellipse

In order to obtain the simplest equation for an ellipse, we place the ellipse at standard position. An ellipse is said to be at standard position when its center is at the origin and its major axis lies on either the ***x***-axis or **y**-axis.

I. Equation of an ellipse at standard position:

There are two possible situations, namely, when the major axis lies on x-axis (called horizontal ellipse) and when the major axis lies on y-axis (called vertical ellipse). We first consider a horizontal ellipse as in Figure 4.19

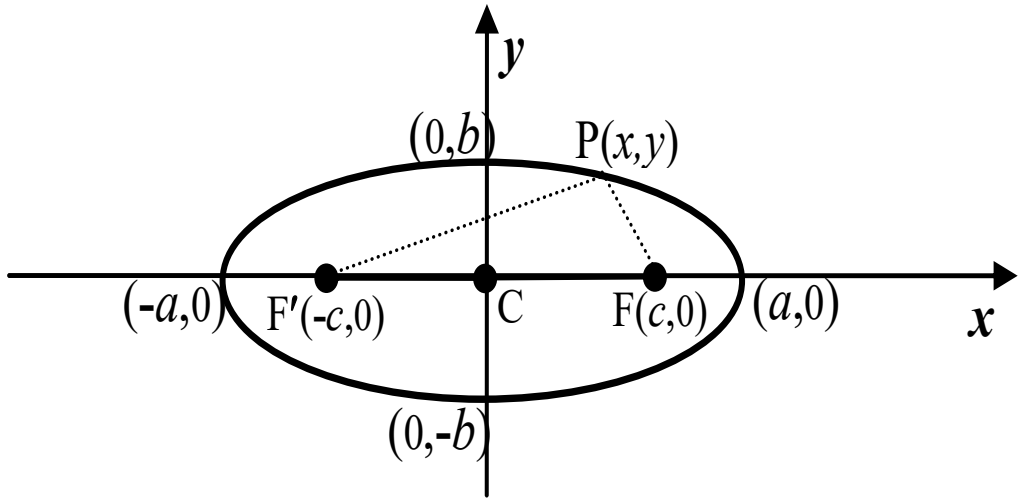


Figure 4.19: Horizontal ellipse at standard position

Let the center of the ellipse be at the origin, C(0,0) and foci at F′(−c,0), F (c,0) and vertices at (−a,0) and (a,0) (see Figure 4.19). Then, a point P(x,y) is on the ellipse iff

$$|PF'| + |PF| = 2a\,.$$

That is, $\sqrt{(x+c)^2+y^2}+\sqrt{(x-c)^2+y^2}=2a$

or $\sqrt{(x-c)^2+y^2}=2a-\sqrt{(x+c)^2+y^2}$

Squaring both sides we get

$$x^2-2cx+c^2+y^2=4a^2-4a\sqrt{(x+c)^2+y^2}+x^2+2cx+c^2+y^2$$

which simplifies to $a\sqrt{(x+c)^2+y^2}=a^2+cx$

Again squaring both sides, we get $a^2(x^2+2cx+c^2)=a^4+2a^2cx+c^2x^2$

which becomes $(a^2-c^2)x^2+a^2y^2=a^2(a^2-c^2)$

Now recalling that $b^2=a^2-c^2$ and dividing both sides by a^2b^2, the equation becomes

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

(Equation of horizontal ellipse at **standard position,** vertices (±a, 0), foci (±c, 0), where $c^2=a^2-b^2$)

For a vertical ellipse at standard position, the same procedure gives the equation

$$\frac{x^2}{b^2}+\frac{y^2}{a^2}=1$$

(Equation of vertical ellipse at **standard position,** vertices (0, ±a), foci (0, ±c), where $c^2=a^2-b^2$)

Example 4.18: Locate the vertices and foci of $16x^2 + 9y^2 = 144$ and sketch its graph.

Solution: Dividing both sides of the equation by 144, we get:

$$\frac{x^2}{9} + \frac{y^2}{16} = 1 \quad \text{or} \quad \frac{x^2}{3^2} + \frac{y^2}{4^2} = 1.$$

This is equation of a vertical ellipse at standard position with a=4, so vertices at (0, ±4), and b=3; i.e., endpoints of the minor axis at (±3, 0). Since $c^2 = a^2 - b^2 = 7 \Rightarrow c = \sqrt{7}$, the foci are $\left(0, \pm\sqrt{7}\right)$. The graph is sketched in Figure 4.20.

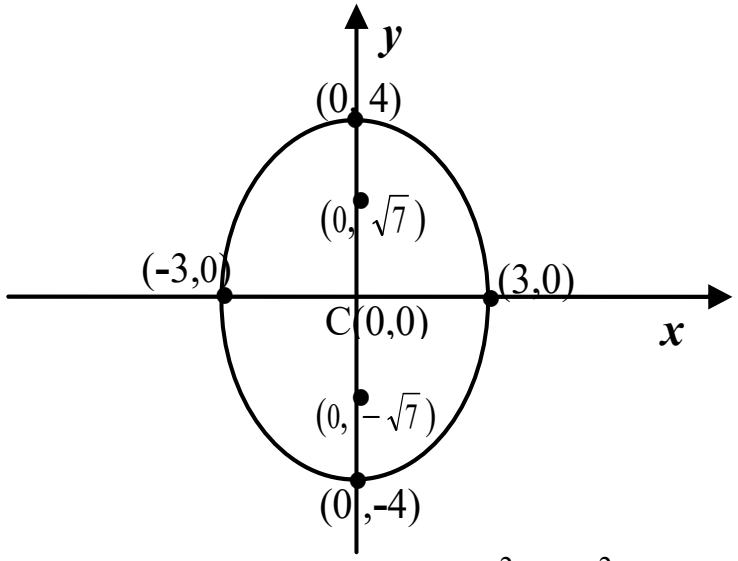


Figure 4.20: $16x^2 + 9y^2$

(II) Equation of shifted Ellipses:

When an ellipse is not at standard position but with center at a point C(h,k), then we can still obtain its equation by considering translation of the xy-axes in such a way that its origin translated to the point C(h,k). This result in a new $X'Y'$ coordinate system whose origin O′ is at C(h,k) so that the ellipse is at standard position relative to the $X'Y'$ system(see, Figure 4.21)

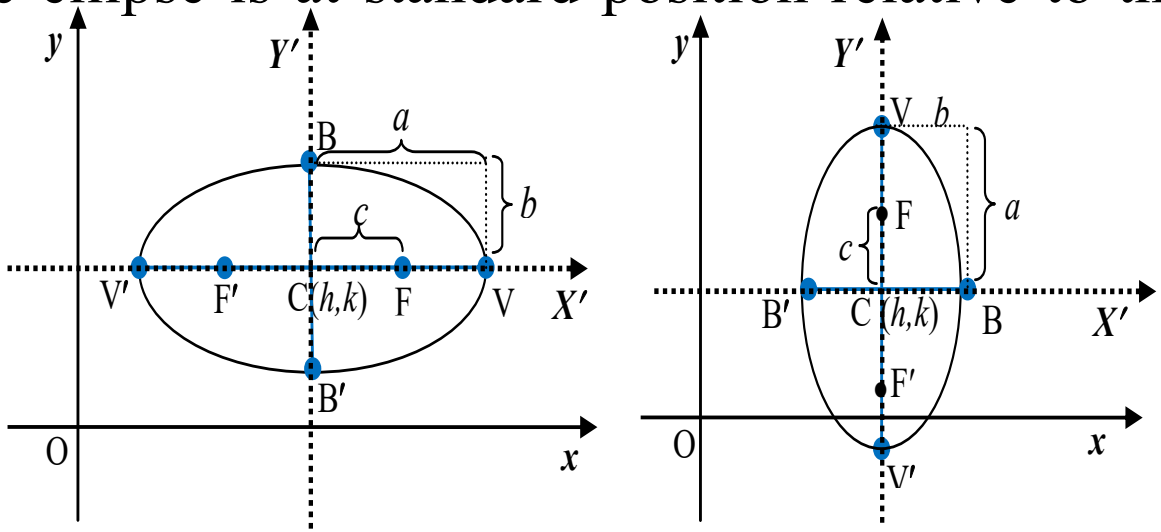


Fig. 4.21: (a) horizontal ellipse, center C(h,k) (b) vertical ellipse, center C(h,k)

Consequently, the equation of the horizontal and vertical ellipses relative to the new $X'Y'$ coordinate system with (x', y') coordinate points are

$$\frac{x'^2}{a^2}+\frac{y'^2}{b^2}=1 \quad \text{and} \quad \frac{x'^2}{b^2}+\frac{y'^2}{a^2}=1, \text{ respectively} \quad \ldots\ldots\ldots\ldots (I).$$

Since the origin of the new coordinate system is at the point (h,k) of the xy-coordinate system, the relationship between a point (x,y) of the ***xy***-coordinate system and (x',y') of the new coordinate system is given by $(x, y) = (x',y') + (h, k)$. That is,

$$x' = x - h, \quad \text{and} \quad y' = y - k.$$

Thus, in the original xy-coordinate system the equations of the horizontal and vertical ellipses with center $C(h, k)$, lengths of major axis $= 2a$ and minor axis $= 2b$ are, respectively, given by

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

(Standard equation of horizontal ellipse with center $C(h,k)$)

and

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$$

(Standard equation of vertical ellipse with center $C(h,k)$)

Example 4.19: The endpoints of the major axis of an ellipse are at $(-3,4)$ and $(7,4)$ and its eccentricity is 0.6. Find the equation of the ellipse and its foci.

Solution: The given vertices are at $V'(-3,4)$ and $V(7,4)$ implies that $2a = |V'V| = 10 \Rightarrow a = 5$; and the center $C(h,k)$ is the midpoint of $V'V \Rightarrow (h,k) = (\frac{-3+7}{2}, \frac{4+4}{2}) = (2,4)$. Moreover, eccentricity $= c/a = 0.6 \Rightarrow c = 5\times 0.6 = 3$. Hence, $b^2 = a^2 - c^2 = 25-9 = 16$. Note that the major axis V'V is horizontal. Therefore, using the standard equation of a horizontal ellipse, the equation of the ellipse is $\dfrac{(x-2)^2}{25} + \dfrac{(y-4)^2}{16} = 1$.

Now, as the center $(h,k) = (2,4)$, $c=3$ and $V'V$ is horizontal, the foci are at $(h \pm c, k) = (2 \pm 3, 4)$.

That is, the foci are at $F'(-1, 4)$ and $F(5, 4)$.

Moreover, the endpoints of major axis are at

$(h, k\pm b) = (2, 4\pm 4) \Rightarrow B'=(2,0)$ and $B=(2,8)$.

The graph of the ellipse is sketched in Figure 4.22.

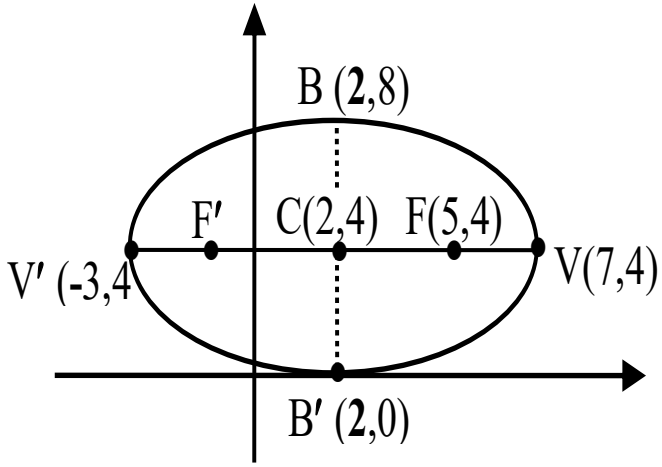


Figure 4.22

Example 4.20: Find the center, foci and vertices of $4x^2 + y^2 + 8x = 0$ and sketch its graph

Solution: Group the x-terms of the equation and complete the square:

$$4(x^2 + 2x) + y^2 = 0$$

$$\Rightarrow 4(x^2+2x+1) + y^2 = 4 \quad \text{(divide both sides by 4)}$$

$$\Rightarrow (x+1)^2 + \frac{y^2}{4} = 1$$

This is equation of a vertical ellipse (major axis parallel to the y-axis), center C=(h,k) = (–1,0),

$a=2,\ b=1. \Rightarrow c^2 = a^2 - b^2 = 4-1 \ \Rightarrow c = \sqrt{3}$

Thus, foci : $F'(-1, -\sqrt{3})$ and $F(-1, \sqrt{3})$,

Vertices: $V = (-1, 2),\ V' = (-1, -2)$;

Endpoints of minor axis: $B=(0,0),\ B'=(-2,0)$;

The graph of the ellipse is sketched in Figure 4.23.

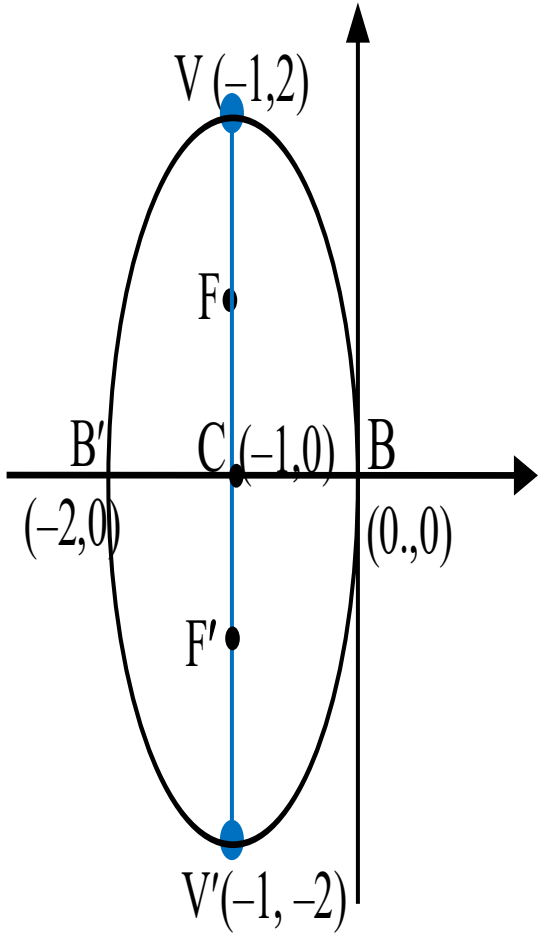


Figure 4.23: $4x^2 + y^2 + 8x = 0$

Remark: Consider the equation: $Ax^2 + Cy^2 + Dx + Ey + F = 0$,

when A and C have the same sign. So, without lose of generality, let $A > 0$ and $C > 0$.

By completing the squares you can show that this equation is equivalent to

$$A\left(x + \frac{D}{2A}\right)^2 + C\left(y + \frac{E}{2C}\right)^2 = \frac{D^2C + E^2A - 4ACF}{4AC}.$$

From this you can conclude that the given equation represents:-

- an ellipse with center $\left(-\frac{D}{2A}, -\frac{E}{2C}\right)$ if $D^2C + E^2A - 4ACF > 0$.
- If $D^2C + E^2A - 4ACF = 0$, the equation is satisfied by the point $\left(-\frac{D}{2A}, -\frac{E}{2C}\right)$ only. In this case, the locus of the equation is called a **point-ellipse** (degenerate ellipse).
- If $D^2C + E^2A - 4ACF < 0$, then the equation has no locus.

Use the information provided to write the standard form equation of each ellipse.

a) $9x^2 + y^2 + 72x - 8y - 176 = 0$

b) vertices: $(-10, 5), (-10, -5)$ and Foci: $(-10, 4), (-10, -4)$

c) Foci:$(-3, 1), (-3, -5)$ and End points of major axis $(-3, 3), (-3, -7)$.

d) Foci:$(-1, -5), (-1, -15)$ and End points of minor axis $(11, -10), (-13, -10)$.

e) Center:$(3, 10)$, Vertex:$(3, -3)$, Focus:$(3, 5)$.

f) End points of major axis :$(16, 10), (-8, 10)$
End points of minor axis :$(4, 20), (4, 0)$

$$16x^2 + 9y^2 - 32x + 36y - 92 = 0$$

4.5 Hyperbolas

4.5.1 Definition of a hyperbola

Definition 4.5: Let F and F′ be two fixed points in the plane. A **hyperbola** is the set of all points in the plane such that the difference of the distance of each point from F and F′ is constant. We shall denote the constant by $2a$, for some $a > 0$. That is, a point P is on the hyperbola if and only if $|PF'| - |PF| = 2a$ (or $|PF| - |PF'| = 2a$, whichever is positive). The two fixed points F and F′ are called the foci of the hyperbola.

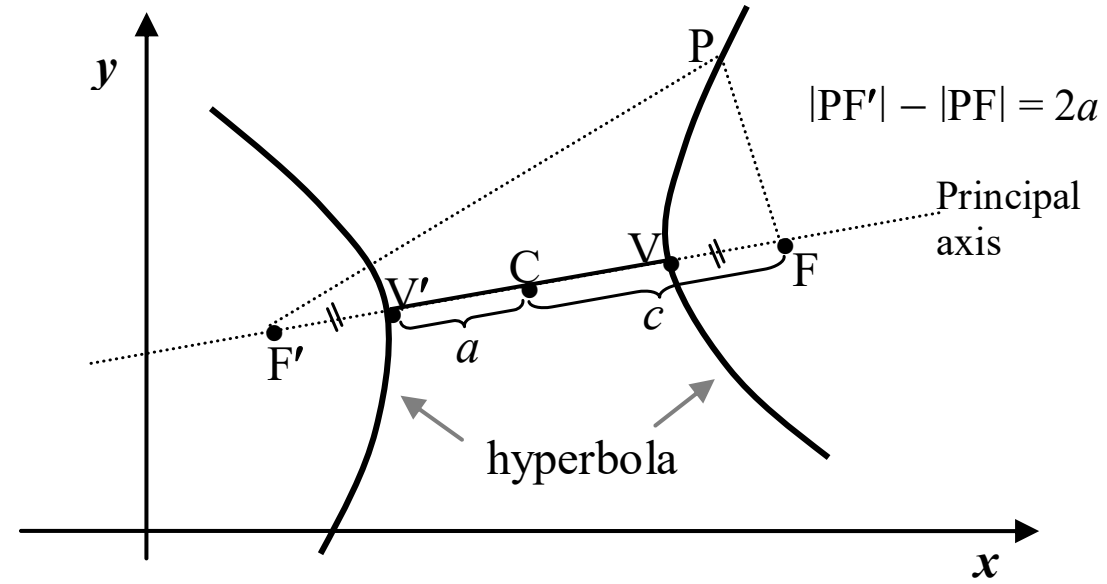


Figure 4.24: Hyperbola

Figure 4.24 illustrates the definition of hyperbola. Notice that the definition of hyperbola is similar to that of an ellipse, the only change is that the sum of distances has become the difference of distances. Here, for the difference of any two unequal values, we take the higher value minus the smaller so that $a > 0$ in the definition.
The following terminologies, notations and relationships are also important with regard to a hyperbola.

- The line through the two foci F′ and F is called the **principal axis** of the hyperbola.
- The point on the principal axis at halfway between the two foci, that is, the midpoint of F′F, is called the **center** of the hyperbola denoted by C.
- We denote the distance between the two foci by $2c$. That is, $|F'F|=2c$ or $|CF|=c = |CF'|$. Note also that $|PF'|< |F'F|+|PF|$ in $\Delta PF'F$ and $|PF'| - |PF| = 2a$, you can show that $a < c$.

- The points V' and V where the hyperbola crosses the principal axis are called **vertices** of the hyperbola. The line segment $V'V$ is called the **transverse axis** of the hyperbola. So, as V' and V are on the hyperbola, the definition requires that $|V'F| - |V'F'| = |VF'| - |VF|$. From this, you can obtain that $|V'F'| = |VF|$. Consequently,

  (i) C is the midpoint of also $V'V$; that is, $|CV'| = |CV|$.

  (ii) $|V'V| = |V'F| - |VF| = |V'F| - |V'F'| = 2a$. (The length of the transverse axis is $2a$)

  (iii) $|V'C| = a = |CV|$ (This follows from (i) and (ii).)

- The **eccentricity** e of a hyperbola is defined to be the ratio of the distance between its foci to the length of its transverse axis. That is, similar to the definition of eccentricity of an ellipse, the eccentricity of a hyperbola is

$$e = \frac{|F'F|}{|V'V|} = \frac{c}{a} \qquad \text{(But here, } e > 1 \text{ because } c > a\text{)}$$

4.5.2 Equation of a hyperbola

First we consider the equation of a standard hyperbola with center at origin. A standard hyperbola is the one whose principal axis (or transverse axis) is parallel to either of the coordinate axes.

I. Equation of a standard hyperbola with center at origin.

If the transverse axis lies on x-axis (called horizontal hyperbola) and when the transverse axis lies on y-axis (called vertical hyperbola). We first consider a horizontal hyperbola with center C(0, 0), vertices V′(–a, 0),V (a, 0) and foci F′(–c, 0), F (c, 0).

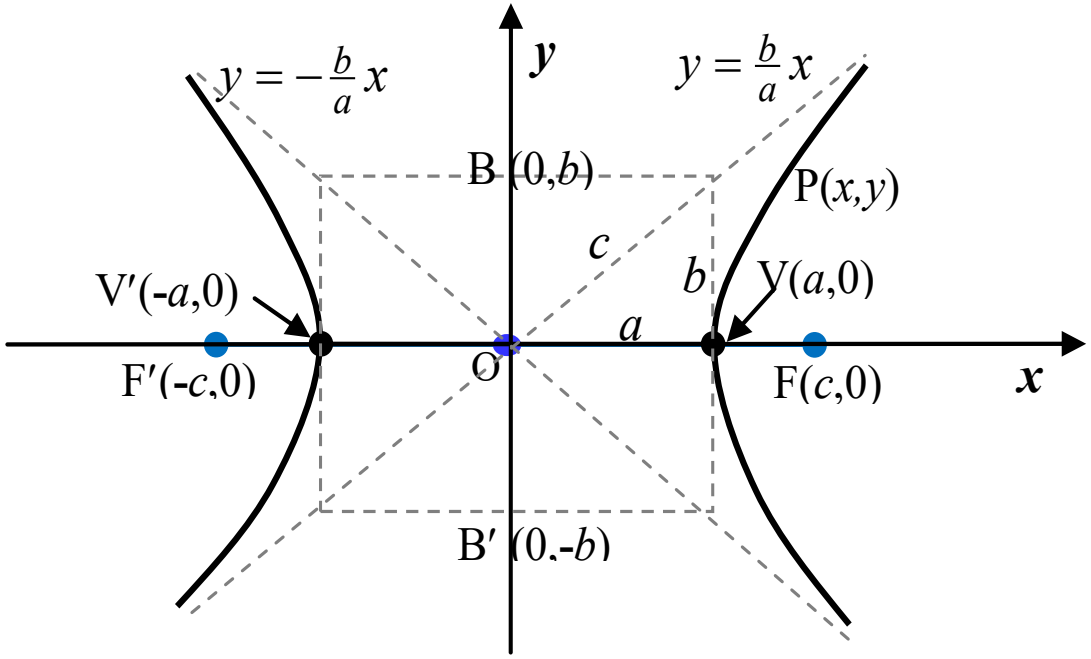


Figure 4.25: Horizontal hyperbola centered at origin

Notice that $c^2 - a^2 > 0$ as $c > a$. Hence, we can put $b^2 = c^2 - a^2$ for some positive b. That is, $a^2 + b^2 = c^2$ so that a, b, c are sides of a right triangle (see, Figure 4.25). The line segment BB′ perpendicular to the transverse axis at C and with endpoints B(0,b) and B′(0,–b) is called **conjugate axis** of the hyperbola. Observe that the midpoint of the conjugate axis is C and its length is $|BB'| = 2b$**.** (b will play important role in equation of the hyperbola and its graph).

Now, for any point P(x,y) on the hyperbola it holds that $|PF'| - |PF| = 2a$ **.**

That is, $\sqrt{(x+c)^2 + y^2} - \sqrt{(x-c)^2 + y^2} = 2a$

or $\sqrt{(x+c)^2 + y^2} = 2a + \sqrt{(x-c)^2 + y^2}$

Squaring both sides we get

$$x^2 + 2cx + c^2 + y^2 = 4a^2 + 4a\sqrt{(x-c)^2 + y^2} + x^2 - 2cx + c^2 + y^2$$

which simplifies to $a\sqrt{(x-c)^2 + y^2} = cx - a^2$

Again squaring both sides and rearranging, we get $(c^2-a^2)x^2-a^2y^2=a^2(c^2-a^2)$.

Recall that we set $b^2=c^2-a^2$. So, using this in the above equation and dividing both sides by a^2b^2, the equation becomes

$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$$

(Equation of horizontal hyperbola with center C(0,0), vertices $(\pm a, 0)$, foci $(\pm c, 0)$, where $c^2=a^2+b^2$)

Note that this hyperbola has no y-intercept because if $x = 0$, then $-y^2 = b^2$ which is not possible. The hyperbola is symmetric with respect to both x-and y- axes.

Also, from this equation we get

$$\frac{x^2}{a^2}=1+\frac{y^2}{b^2}\geq 1 \quad \text{implies that} \quad x^2\geq a^2. \quad \text{So,} \quad |x|=\sqrt{x^2}\geq\sqrt{a^2}=a.$$

Therefore, we have $x \geq a$ or $x \leq -a$. This means that the hyperbola consists of two parts, called its *branches*. Moreover, if we solve for y from the equation we get

$$y = \pm\frac{b}{a}\sqrt{x^2 - a^2} \to \pm\frac{b}{a}x \text{ as } x \to \infty.$$

This means the hyperbola will approach (but never reaches) the line $y = \pm\frac{b}{a}x$ as x gets larger and larger. That is, the lines $y = \pm\frac{b}{a}x$ are the **asymptotes** of the hyperbola.

In sketching a hyperbola, it is best to draw the rectangle formed by the line $y = \pm b$ and $x = \pm a$ and the to draw the asymptotes which are along the diagonals of the rectangle (as shown by the dashed lines in Figure 4.25). The hyperbola lies outside the rectangle and inside the asymptotes. It opens around the foci.

Example 4.21: Find an equation of the hyperbola whose foci are

$F'(-5, 0)$ and $F(5, 0)$ and contains point $P(5, 16/3)$.

Solution: It is horizontal hyperbola with center $(0,0)$ and $c = 5$.

In addition, as $P(5, 16/3)$ is on the hyperbola we have that

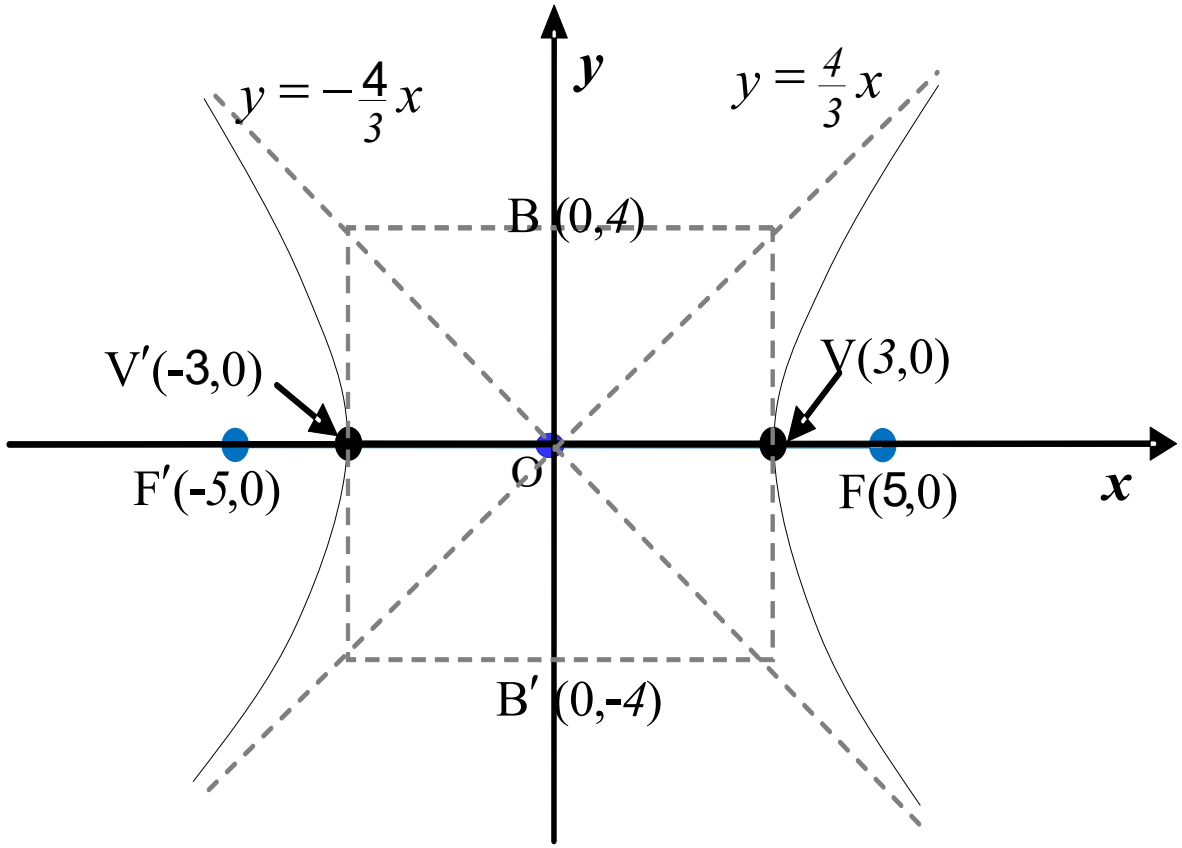
$|PF'| - |PF| = 2a$. That is, $\sqrt{(5+5)^2 + \left(\frac{16}{3}\right)^2} - \sqrt{(5-5)^2 + \left(\frac{16}{3}\right)^2} = 2a$

$\Rightarrow \quad a = 3$. (So, its vertices are $(-3, 0)$ and $(3, 0)$).

Now, using the relationship $b^2 = c^2 - a^2$, we get $b^2 = 25 - 9 = 16$.

Therefore, the equation of the hyperbola is $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

Thus the lines $y = \pm \frac{4}{3}x$ are the **asymptotes** of the hyperbola.



For a vertical hyperbola with center at origin (i.e., when transverse axis lies on y-axis), by reversing the role of x and y we obtain the following equation which is illustrated in Figure 4.26.

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

(Equation of vertical hyperbola with center C(0,0), foci $(0, \pm c)$, vertices $(0, \pm a)$, where $c^2 = a^2 + b^2$ and asymptotes $y = \pm(a/b)x$)

Note:
- For a vertical hyperbola, the coefficient of y^2 is positive and that of x^2 is negative .
- a^2 is always the denominator of the positive term.

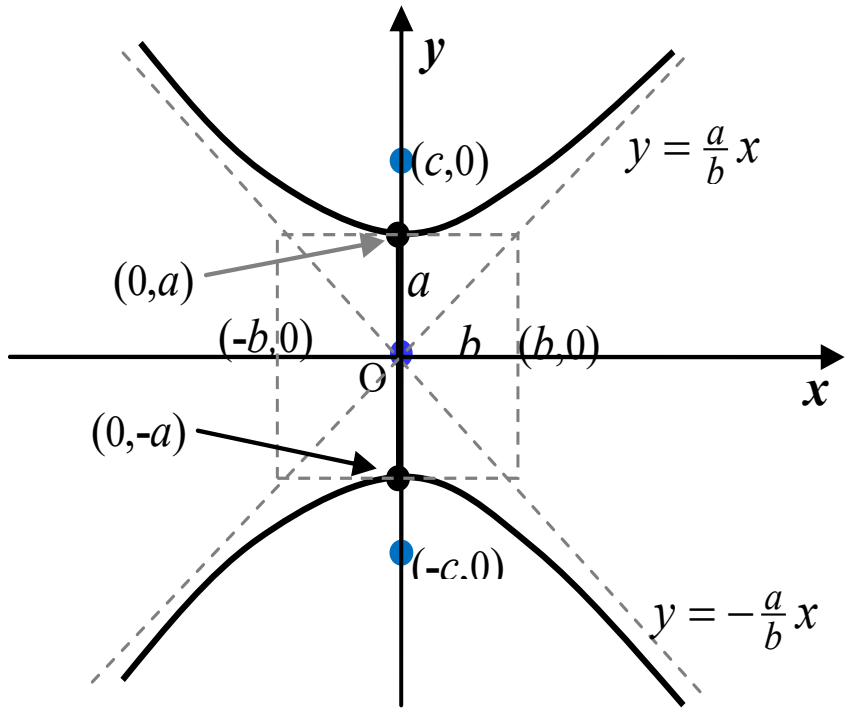


Figure 4.26: Vertical hyperbola centered at origin

Example 4.22: Find the foci and equation of the hyperbola with vertices $V'(0,-1)$ and $V(0, 1)$ and an asymptote $y = 2x$.

Solution: It is a vertical hyperbola with center $C(0,0)$ and $a = |CV| = 1$. Since an asymptote of such vertical hyperbola is $y = (a/b)x$ and the slope of the given asymptote is 2, we have $a/b = 2$ $\Rightarrow$ $1/b = 2 \Rightarrow b = 1/2$. Thus, $c^2 = a^2 + b^2 = 1 + 1/4 = 5/4$.

So, the foci are $\left(0, \pm \sqrt{5}/2\right)$ and the equation of the hyperbola is $y^2 - 4x^2 = 1$.

(You may sketch the hyperbola)

(II) Equation of shifted hyperbolas:

In this case, we form the equation of the hyperbolas by using the translation of the xy-coordinate system that shifts its origin to the point $C(h, k)$. As discussed in Section 4.4, the effect of this translation is just replacing x and y by $x-h$ and $y-k$, respectively, in the equation of the desired hyperbola.

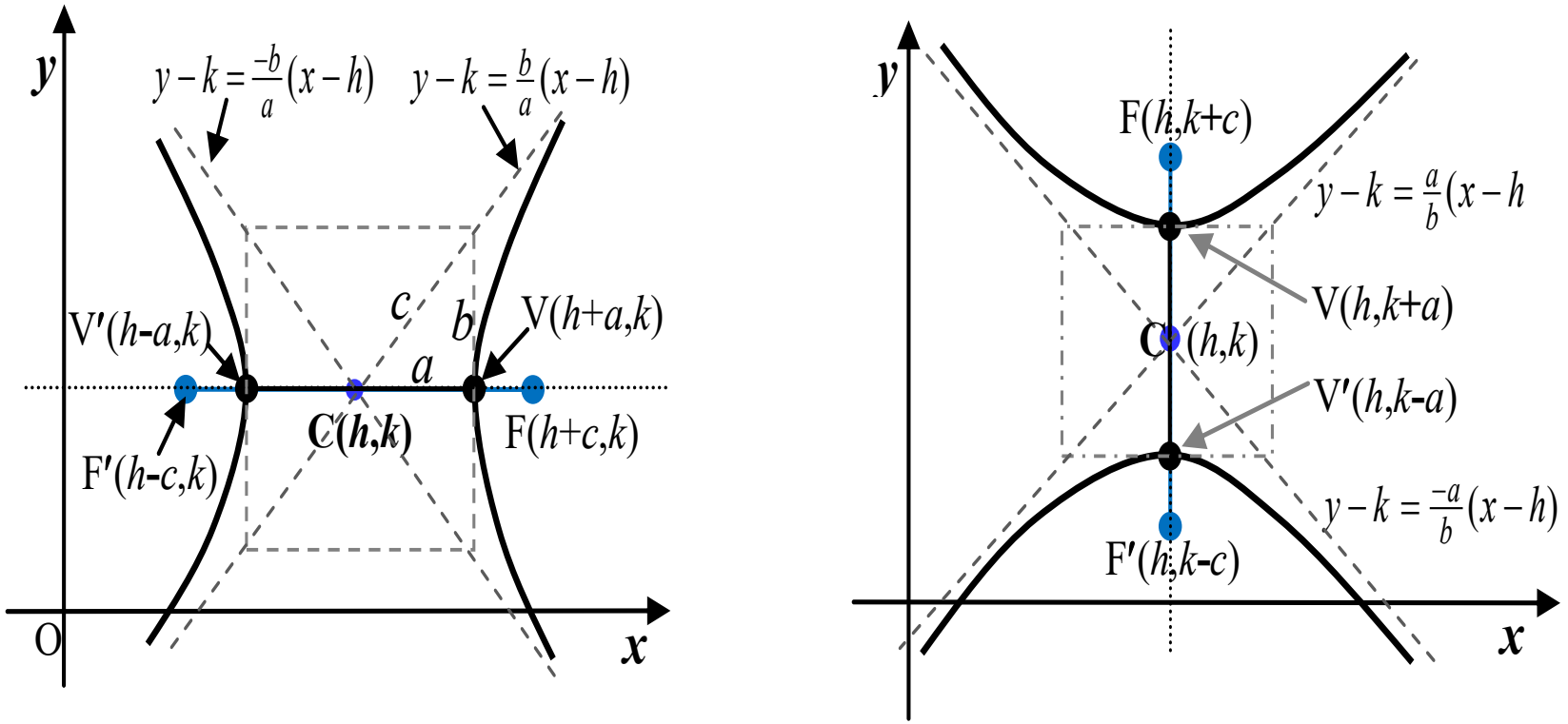


Fig 4.27: (a) Horizontal hyperbola with center $C(h,k)$ (b) Vertical hyperbola with center $C(h,k)$

Therefore, the standard equation of a horizontal hyperbola (transverse axis parallel to x-axis) with center C(h,k), length of transverse axis $=2a$, and length of conjugate axis $=2b$ is

$$\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$$

Center: C(h,k),

Vertices: V′ ($h-a,k$), V($h+a, k$),

Foci : F′($h-c, k$), F($h+c, k$), where $c^2=a^2+b^2$

Asymptotes: $y-k=\pm\frac{b}{a}(x-h)$

Similarly, the standard equation of a vertical hyperbola (transverse axis parallel to y-axis) with center C(h,k), length of transverse axis $=2a$, and length of conjugate axis $=2b$ is

$$\frac{(y-k)^2}{a^2}-\frac{(x-h)^2}{b^2}=1$$

Center: C(h,k),

Vertices: V′ ($h,k-a$), V($h, k+a$),

Foci : F′($h, k-c$), F($h, k+c$), where $c^2=a^2+b^2$

Asymptotes: $y-k=\pm\frac{a}{b}(x-h)$

Example 4.23: Find the foci, vertices and the asymptotes of the hyperbola whose equation is

$4(x+1)^2-(y-2)^2=4$

and sketch the hyperbola.

Solution: Dividing both sides of the equation by 4 yields

$$(x+1)^2-\frac{(y-2)^2}{4}=1$$

This is equation of a hyperbola with center C(–1, 2). Note that the 'x^2-term' is positive indicates that the hyperbola is horizontal (principal axis y=2), a=1, b=2, and $c^2=a^2+b^2 \Rightarrow c=\sqrt{5}$. As a result the foci are at $\left(-1-\sqrt{5},2\right)$ and $\left(-1+\sqrt{5},2\right)$, vertices are at (–2,2) and (0,2) and the asymptotes are the lines $y-2=\pm 2(x+1)$, that is, $y=2x+4$ and $y=-2x$. Consequently, the hyperbola is sketched as in Figure 4.28.

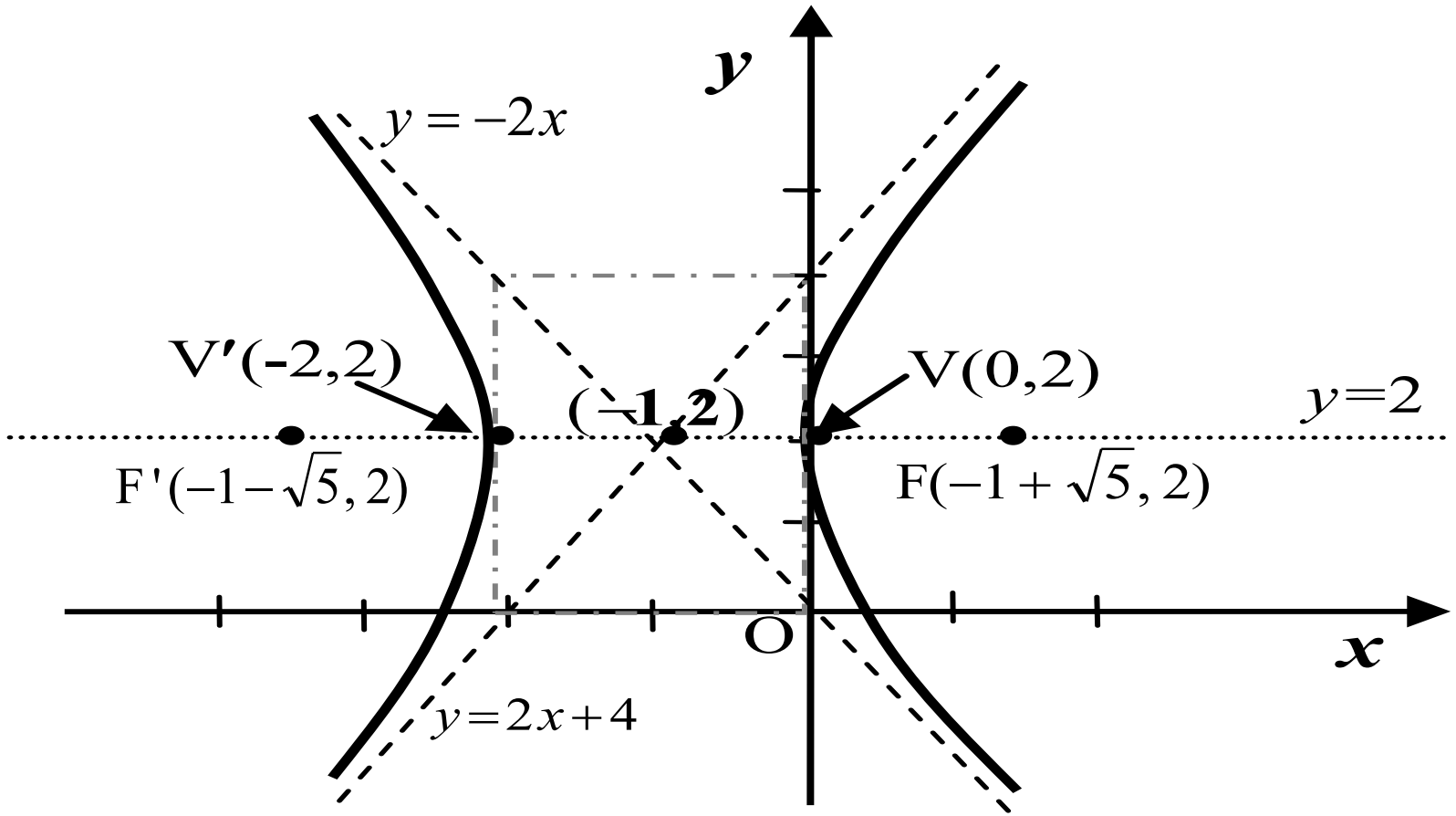


Figure 4.28: $4(x+1)^2-(y-2)^2=4$

Example 4.24: Find the foci of the conic $9x^2-4y^2-72x+8y+176=0$ and sketch its graph.

Solution: Group the x-terms and y-terms of the equation and complete their squares:

$$\Rightarrow\ 9x^2-72x-4y^2+8y=-176 \qquad \text{(Multiply both sides by } -1\text{)}$$

$$\Rightarrow\ -9x^2+72x+4y^2-8y=176$$

$$\Rightarrow\ 4(y^2-2y)-\ 9(x^2-8x)=176$$

$$\Rightarrow\ 4(y^2-2y+\mathbf{1}^2)-\ 9(x^2-8x+\mathbf{4}^2)=176+\mathbf{4}-\mathbf{144}$$

$$\Rightarrow\ 4(y-1)^2-\ 9(x-4)^2=36 \quad \text{(Next, divide each by 36)}$$

$$\Rightarrow\ \frac{(y-1)^2}{9}-\ \frac{(x-4)^2}{4}=1$$

This is standard equation of a hyperbola whose transverse axis is parallel to the y-axis (as its 'y^2 term' is positive) with center C (4,1), a^2=9 and b^2=4 . $\Rightarrow\ c^2=a^2+b^2=13\ \Rightarrow c=\sqrt{13}$.

Thus, foci are $F'(4,1-\sqrt{13})$ and $F(4,1+\sqrt{13})$, and vertices $(4,\ 1\pm 3)$, i.e.,, $V'(4,-2)$ and $V(4,\ 4)$. Moreover, the asymptotes are $y-k=\pm\frac{a}{b}(x-h)$. Hence, the asymptotes are l_1: $y-1=\frac{3}{2}(x-4)$ and l_2: $y-1=-\frac{3}{2}(x-4)$. The hyperbola is sketched in Figure 4.29

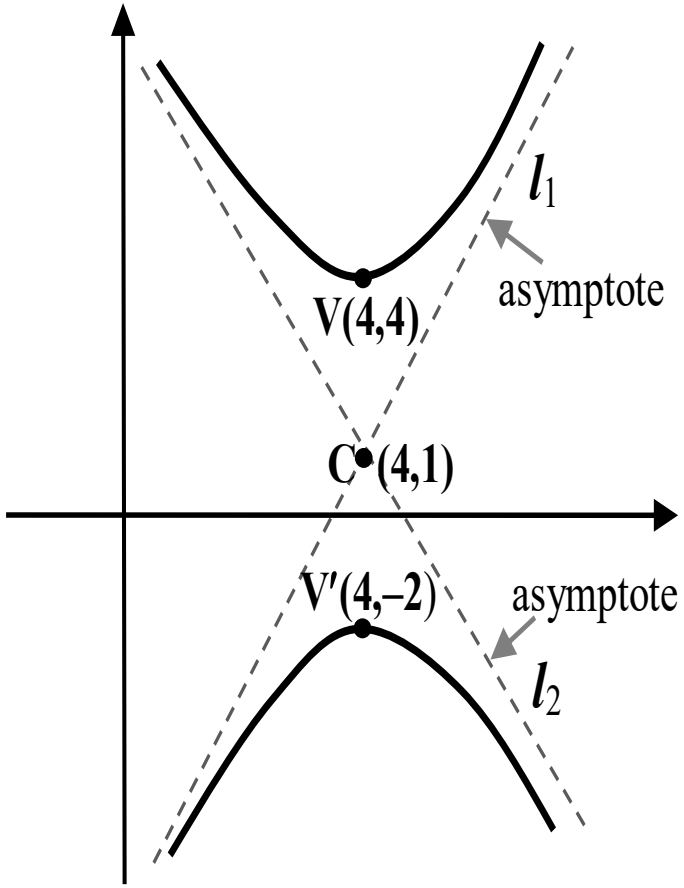


Figure 4.29: $\frac{(y-1)^2}{9}-\frac{(x-4)^2}{4}=1$

Example 4.25: Determine the locus or type of the conic section given by the equation

$$-x^2 + y^2 + 4x - 2y = 3.$$

Solution: Grouping the x-terms and y-terms of the equation and completing their squares yield

$$(y-1)^2 - (x-2)^2 = 0$$

$$\Rightarrow \quad (y-1)^2 = (x-2)^2$$

$$\Rightarrow \quad y - 1 = \pm\sqrt{(x-2)^2} = \pm(x-2)$$

This represents pair of two lines intersecting at (2, 1), namely, $y = x-1$ and $y = -x+3$.

Remark: Consider the equation: $Ax^2 + Cy^2 + Dx + Ey + F = 0$ when $AC < 0$;

(i.e., A and C have opposite signs). Then, by completing the squares of x-terms and y-terms you can convert the equation to the following form:

$$A\left(x + \frac{D}{2A}\right)^2 + C\left(y + \frac{E}{2C}\right)^2 = \frac{D^2C + E^2A - 4ACF}{4AC}.$$

Now, letting $\Delta = D^2C + E^2A - 4ACF$, you can conclude the following:

- If $\Delta \neq 0$, the equation represents a hyperbola with center $\left(-\frac{D}{2A}, -\frac{E}{2C}\right)$.

If $\Delta = 0$, the equation becomes $y + \frac{E}{2C} = \pm\sqrt{\left|\frac{A}{C}\right|}\left(x + \frac{D}{2A}\right)$ which are two lines intersecting at $\left(-\frac{D}{2A}, -\frac{E}{2C}\right)$. In this case, it is called **degenerate** hyperbola.

4. 6 The General Second Degree Equation (Reading Assignment)

4 .6.1 Rotation of Coordinate Axes

A rotation of the x and y coordinate axes by an angle θ about the origin O(0,0) creates a new $x'y'$-coordinate system whose x'-axis is the line obtained by rotating the x-axis by angle θ about O and y'-axis is the line obtained by rotating the y-axis in the same way. This makes a point P to have two sets of coordinates denoted by (x,y) and (x',y') relative to the xy- and $x'y'$-coordinate axes, respectively. (See Figure 4.30).

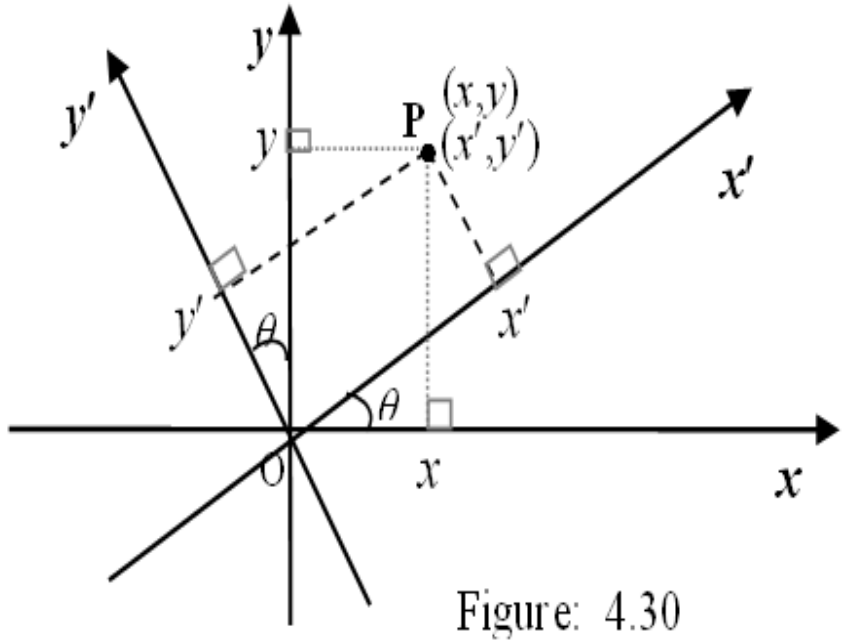


Figure: 4.30

The angle θ considered in the above discussion is called the angle of rotation. Our aim is to find the relationships between the coordinates (x,y) and the coordinates (x',y') of the same point P.

To find this relationships, let P(x,y) be any point in xy-plane, θ be an angle of rotation (i.e., θ is angle between x and x' axes) and ϕ be the angle between OP and $\boldsymbol{x'}$-axis (See Figure 4.31).
So, letting $|OP| = r$ observe that

$$x' = r\cos\phi, \qquad y' = r\sin\phi \quad \ldots\ldots\ldots \quad (1)$$

and

$$x = r\cos(\theta+\phi), \qquad y = r\sin(\theta+\phi) \quad \ldots\ldots \quad (2)$$

Then, using the trigonometric identities

$$\cos(\theta+\phi) = \cos\theta\cos\phi - \sin\theta\sin\phi$$
$$\sin(\theta+\phi) = \sin\theta\cos\phi + \cos\theta\sin\phi$$

and (1), the equations in (2) become

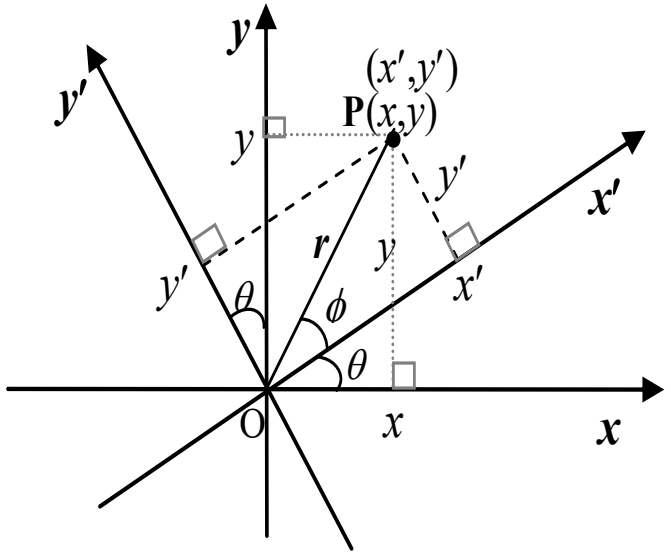


Figure: 4.31

$$x = x'\cos\theta - y'\sin\theta$$
$$y = x'\sin\theta + y'\cos\theta \quad \ldots \quad (3)$$

Moreover, these equations can be solved for x' and y' in terms of x and y to obtain

$$x' = x\cos\theta + y\sin\theta$$
$$y' = -x\sin\theta + y\cos\theta \quad \ldots \quad (4)$$

The Equations (3) and (4) are called **rotation formulas**. It follows that if the angle of rotation θ is given, then Equation (3) can be used to determine the x and y coordinates of a point P if we know its x' and y' coordinates. Similarly, Equation (4) can be used to determine the x' and y' coordinates of P if we know its x and y coordinates.

Example 4.26: Suppose the ***x*** and ***y*** coordinate axes are rotated by $\pi/4$ about the origin.

(a) Find the coordinates of P(1, 2) relative to the new ***x***' and ***y***' axes.

(b) Find the equation of the curve $xy = 1$ relative to the new $x'y'$-coordinate system and sketch its graph.

Solution: The given information about P and the curve are relative to the xy-coordinate system and we need to express them in terms of x' and y' coordinates relative to the new $x'y'$-coordinate system obtained under the rotation of the original axes by $\theta = \pi/4$ rad about the origin. Thus, we use $\cos\frac{\pi}{4} = \sin\frac{\pi}{4} = \frac{\sqrt{2}}{2}$ in the relevant rotation formula to obtain the following.

(a) Since P(1,2) has the coordinates $x=1$ and $y=2$, its x' and y' coordinates are, using formula (4)

$$x' = \frac{\sqrt{2}}{2}(1) + \frac{\sqrt{2}}{2}(2) = \frac{3\sqrt{2}}{2} \quad and \quad y' = -\frac{\sqrt{2}}{2}(1) + \frac{\sqrt{2}}{2}(2) = \frac{\sqrt{2}}{2}$$

Therefore, the coordinates of P relative to the new ***x***' and ***y***' axes are $\left(\frac{3\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$.

(b) We need to express x and y in the equation $xy = 1$ in terms of x' and y' using the rotation formula (3). So, again since $\cos\frac{\pi}{4} = \sin\frac{\pi}{4} = \frac{\sqrt{2}}{2}$, we obtain from formula (3):

$$x = \frac{\sqrt{2}}{2}x' - \frac{\sqrt{2}}{2}y' \quad \text{and} \quad y = \frac{\sqrt{2}}{2}x' + \frac{\sqrt{2}}{2}y'$$

Therefore, $xy = 1 \Rightarrow \left(\frac{\sqrt{2}}{2}x' - \frac{\sqrt{2}}{2}y'\right)\left(\frac{\sqrt{2}}{2}x' + \frac{\sqrt{2}}{2}y'\right) = 1$

$$\Rightarrow \left(\frac{\sqrt{2}}{2}x'\right)^2 - \left(\frac{\sqrt{2}}{2}y'\right)^2 = 1$$

$$\Rightarrow \frac{x'^2}{2} - \frac{y'^2}{2} = 1$$

Note that this is an equation of a hyperbola with center at origin vertices $(-\sqrt{2},0)$ and $(\sqrt{2},0)$ in the $x'y'$-coordinate system with principal axis on $\boldsymbol{x}'$-axis. Since the $\boldsymbol{x}$ and $\mathbf{y}$- axes were rotated though an angle of $\pi/4$ to obtain $\boldsymbol{x}'$ and $\boldsymbol{y}'$-axes, the hyperbola can be sketched as in Figure 4.32. (You may use Formula (3) to show that the vertices $(-\sqrt{2},0)$ and $(\sqrt{2},0)$ are $(-1,-1)$ and $(1,1)$, respectively, relative to the $\boldsymbol{x}$ and $\boldsymbol{y}$-axes).

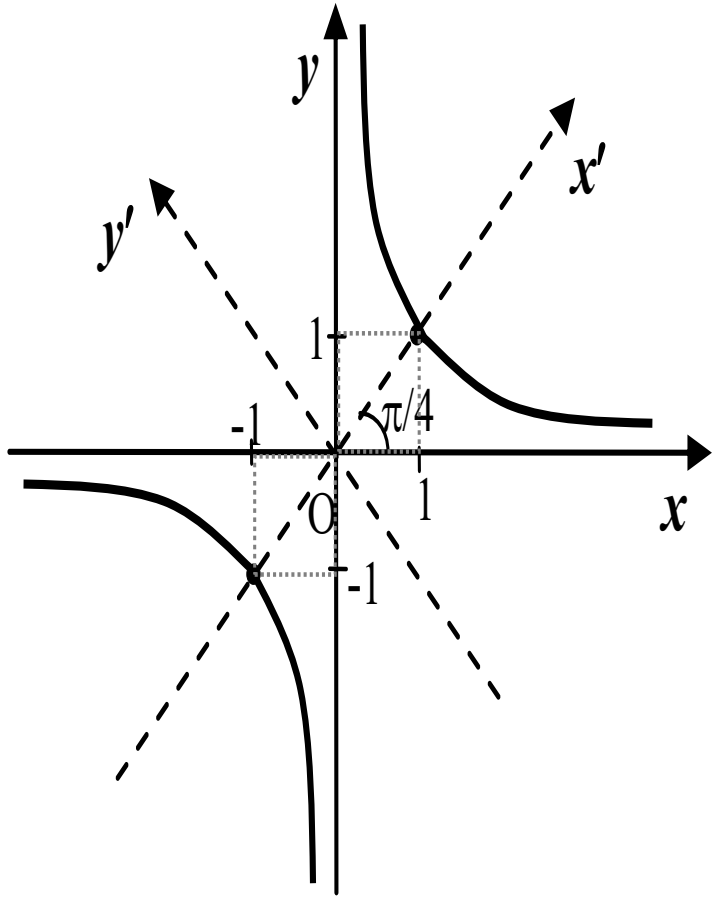


Figure 4.32: $xy = 1$

Example 4.27: Find an equation of the ellipse whose center is the origin, vertices are $(-4,-3)$ and $(4, 3)$, and length of minor axis is 6.

Solution: The position of the ellipse is as shown in Figure 4.33.

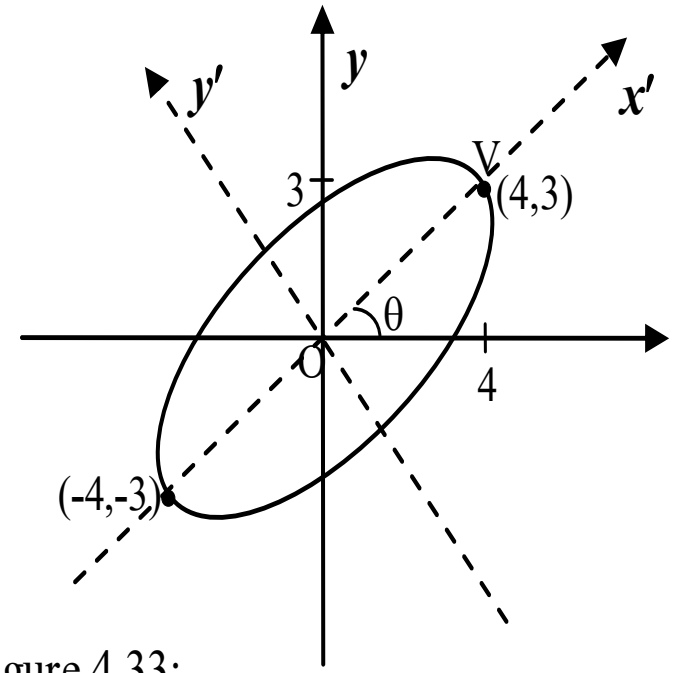


Figure 4.33:

To apply the standard equation of ellipse we use the $x'y'$-coordinate system such that the x'-axis coincide with the major axis of the ellipse. Therefore, the equation of the ellipse relative to the $x'y'$ system is

$$\frac{x'^2}{a^2}+\frac{y'^2}{b'^2}=1.$$

Moreover, from the given information, $a^2 = |\mathrm{OV}|^2 = 3^2+4^2 = 25$; and

length of minor axis $= 2b = 6 \Rightarrow b = 3$. So, $b^2 = 9$.

Hence, the equation of the ellipse relative to the $x'y'$ -coordinate system is

$$\frac{x'^2}{25}+\frac{y'^2}{9}=1 \qquad \text{or} \qquad 9x'^2+25y'^2=225 \ . \ . \ . \ . \ . \ . \ . \ . \ . \ (1)$$

Now we use the rotation formula to express the equation relative to our xy-coordinate system.

So, let θ the angle between $\boldsymbol{x}$-axis and $\boldsymbol{x}'$-axis. Then, observe that

$$\cos\theta = 4/5 \quad \text{and} \quad \sin\theta = 3/5.$$

Thus, using rotation formula (4) we get:

$$x' = x\cos\theta + y\sin\theta = \tfrac{4}{5}x + \tfrac{3}{5}y$$
$$y' = -x\sin\theta + y\cos\theta = \tfrac{-3}{5}x + \tfrac{4}{5}y$$

Now we substitute these for x' and y' in (1) to obtain $9\left(\frac{4}{5}x+\frac{3}{5}y\right)^2+25\left(\frac{-3}{5}x+\frac{4}{5}y\right)^2=225$

And simplifying this we get $369x^2 - 384xy + 481y^2 - 5625 = 0$ which is the equation of the ellipse in the xy- coordinate system.

4.6.2 The General Second Degree Equations

In the previous sections we have seen that the equation of the form

$$Ax^2 + Cy^2 + Dx + Ey + F = 0, \quad \ldots\ldots\ldots\ldots\ldots \quad (1)$$

represents a conic section (a parabola, ellipse or hyperbola) whose axis of symmetry is parallel to one of the coordinate axes except in degenerate cases. In Subsection 4.6.1 we have also seen some examples of conic sections whose equations involve xy term when their lines of symmetry are not parallel to either of the axes. Now we would like to analyze the graph of any quadratic (second degree) equation in x and y of the form

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0, \text{ where } B \neq 0. \quad \ldots\ldots\ldots\ldots\ldots \quad (2)$$

Theorem 4.3: Consider a general second degree equation of the form (2), i..e.,

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0, \quad \text{where } B \neq 0, \quad \ldots\ldots\ldots \quad (2)$$

there is a rotation angle $\theta \in (0, \pi/2)$ through which the xy-coordinate system rotates to a new $x'y'$-coordinate system in which Equation (2) reduces to the form

$$A'x'^2 + C'y'^2 + D'x' + E'y' + F' = 0. \quad \ldots\ldots\ldots \quad (3)$$

Remark: If $A = C$, then $\cot 2\theta = \frac{A-C}{B} = 0 \Rightarrow 2\theta = \pi/2 \Rightarrow \theta = \pi/4$.

Therefore, we can rewrite the result of the above Theorem as follows:

The rotation of the xy-coordinate system by angle θ creates an $x'y'$-coordinate system in which a general second degree equation $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0, \quad B \neq 0,$

is converted to an equation $A'x'^2 + C'y'^2 + D'x' + E'y' + F' = 0$ (with no $x'y'$ term)

if we choose $\theta \in (0, \pi/2)$ such that

$$\tan 2\theta = \frac{B}{A - C}, \quad \text{if } A \neq C$$

$$\theta = \frac{\pi}{4}, \quad \text{if } A = C$$

Example 4.28: Use rotation of axes to eliminate the xy term in each of the following equations, describe the locus (type of conic section) and sketch the graph of the equation

a) $x^2 + 2xy + y^2 - 8\sqrt{2}x + 8\sqrt{2}y - 32 = 0$

(b) $73x^2 - 72xy + 52y^2 + 30x + 40y - 75 = 0$

Solution:

(a) Given: $x^2 + 2xy + y^2 - 8\sqrt{2}x + 8\sqrt{2}y - 32 = 0 \Rightarrow$ A= C=1. So, from the above Remark, the rotation angle is $\theta = \pi/4 \Rightarrow \cos\theta = \sin\theta = \frac{1}{\sqrt{2}}, \Rightarrow x = \frac{x'-y'}{\sqrt{2}}$ and $x = \frac{x'+y'}{\sqrt{2}}$.

Now we substitute these for x and y in the given equation:

$$\left(\frac{x'-y'}{\sqrt{2}}\right)^2 + 2\left(\frac{x'-y'}{\sqrt{2}}\right)\left(\frac{x'+y'}{\sqrt{2}}\right) + \left(\frac{x'+y'}{\sqrt{2}}\right)^2 - 8\sqrt{2}\left(\frac{x'-y'}{\sqrt{2}}\right) + 8\sqrt{2}\left(\frac{x'+y'}{\sqrt{2}}\right) - 32 = 0$$

Expanding the squared expressions, combining like terms and simplifying, we obtain

$$x'^2 + 8y' - 16 = 0 \quad \text{or,} \quad x'^2 = -8(y' - 2)$$

This is an equation of a parabola. Its vertex is $(h',k') = (0,2)$ relative to the $x'y'$-system, principal axis is on y'-axis and open towards negative y' direction. (You can show that its vertex is $(h, k) = \left(-\frac{2}{\sqrt{2}}, \frac{2}{\sqrt{2}}\right)$ relative to the xy-system). The graph of the equation is sketched in Figure 4.34.

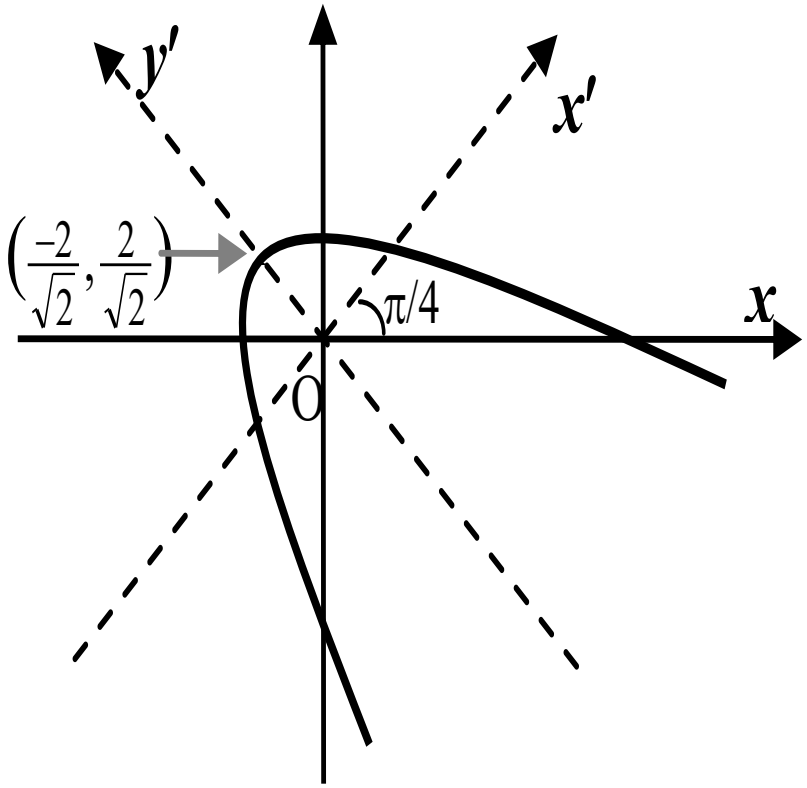


Figure 4.34: $x^2 + 2xy + y^2 - 8\sqrt{2}x + 8\sqrt{2}y - 32 = 0$

(b) Given: $73x^2 - 72xy + 52y^2 + 30x + 40y - 75 = 0 \quad \Rightarrow \quad A = 73, \quad B = -72$ and $C = 52$.

Hence,

$$\tan 2\theta = \frac{B}{A-C} = -\frac{72}{21} = -\frac{24}{7} \Rightarrow \text{The terminal side of } 2\theta \text{ is through } (-7, 24) \text{ since } 0 < 2\theta < \pi.$$

$$\Rightarrow \cos 2\theta = \frac{-7}{25}. \text{ Now as } 0 < \theta < \pi/2, \text{ both } \cos\theta \text{ and } \sin\theta \text{ are positive. Hence,}$$

$$\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}} = \sqrt{\frac{1-7/25}{2}} = \frac{3}{5} \qquad \text{and} \qquad \sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}} = \sqrt{\frac{1+7/25}{2}} = \frac{4}{5}$$

This implies the x'-axis is through the coordinate point (3,4), that is the line $y = (4/3)x$.

Therefore, using the rotation formula (3), we get

$$x = \frac{3x'-4y'}{5} \qquad \text{and} \qquad y = \frac{4x'+3y'}{5}$$

Now we substitute these for x and y in the given equation to obtain

$$\frac{73}{25}(3x'-4y)^2 - \frac{72}{25}(3x'-4y')(4x'+3y') + \frac{52}{25}(4x'+3y')^2 + \frac{30}{5}(3x'-4y') + \frac{40}{5}(4x'+3y') - 75 = 0.$$

Expanding the squared expressions, combining like terms and simplifying, we obtain

$$25x'^2 + 100y'^2 + 50x' - 75 = 0$$

Completing the square for x' terms and divide by 100 to get

$$\frac{(x'+1)^2}{4} + y'^2 = 1$$

which is an ellipse with center at $(h',k')= (-1, 0)$ relative to the $x'y'$-system, major axis on x'-axis (which is the line $y=(4/3)x$), length of major axis =4 and length of minor axis =2. (You can show that the center is $(h, k) = \left(-\frac{3}{5}, -\frac{4}{5}\right)$ relative to the xy-system). The graph of the equation is sketched in Figure 4.35.

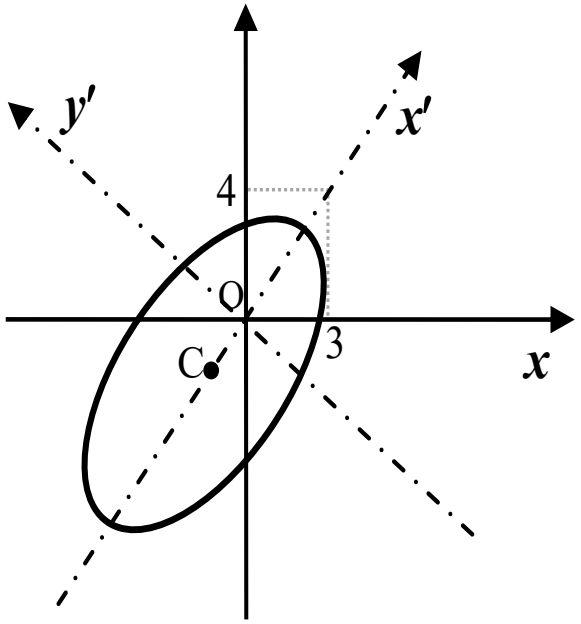


Figure 4.35: $73x^2 - 72xy + 52y^2 + 30x + 40y - 75=0$

THE END

THANK YOU!!!